Financial repression in general equilibrium: The case of the United States, 1948–1974

By Martin Kliem, Alexander Kriwoluzky, Gernot J. Müller, and Alexander Scheer*

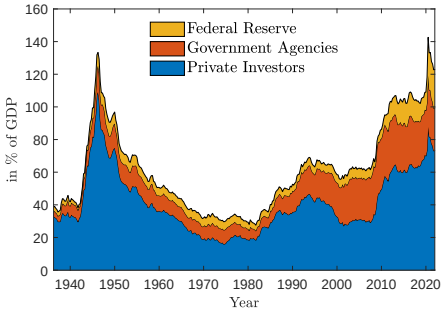
Financial repression lowers the return on government debt and contributes, all else equal, towards its liquidation. However, its full effect on the debt-to-GDP ratio hinges on how repression impacts the economy at large because it alters investment and saving decisions. We develop and estimate a New Keynesian model with financial repression. Based on U.S. data for the period 1948–1974, we find, consistent with earlier work, that repression was pervasive but gradually phased out. A model-based counterfactual shows that phasing out repression lowered—rather than raised—the debt-to-GDP ratio, by 20 percentage points, and raised GDP by 5 percent.

Public debt has recently reached record levels, as Panel A of Figure 1 illustrates for the U.S. based on various measures. There is consensus among policy makers that prudence calls for a reduction of debt, however gradual, in order to create fiscal space and prepare for future crises (IMF, 2023).[1] Classical debt reduction policies include boosting growth, primary surpluses, and inflation, all of which are difficult to engineer but are frequently deployed with varying intensity (Arslanalp and Eichengreen, 2023; Barro and Bianchi, 2026). Also, the same panel shows that the debt-to-GDP ratio declined strongly post-WW2. Next to the above mentioned policies, this was arguably also due to financial repression, which allows the government to borrow at artificially low rates (Reinhart and Sbrancia, 2015; Acalin and Ball, 2026). And while the notion of financial repression was originally developed in the context of developing economies (McKinnon, 1973; Shaw, 1973), Panel B of Figure 1 illustrates that (holding period) returns on U.S. government debt were indeed spectacularly low during and after WW2. This holds not only relative to later periods, but also compared to short-term rates and inflation. Such low returns are also unique in a broader historical context (Hall and Sargent, 2021, 2022).

* This Version: September 2026. Kliem: Deutsche Bundesbank (martin.kliem@bundesbank.de); Kriwoluzky: DIW Berlin and Freie Universität Berlin (akriwoluzky@diw.de); Müller: University of Tübingen, CEPR and CESifo (gernot.mueller@uni-tuebingen.de); Scheer: Deutsche Bundesbank (alexander.scheer@bundesbank.de). We thank Francesco Bianchi (the editor) and three anonymous referees, Julien Acalin, Klaus Adam, Arvind Krishnamurthy, Eric Leeper, Harald Uhlig, Almuth Scholl, and Martin Wolf as well as seminar and workshop participants at various events for very helpful comments and suggestions. This paper represents the authors' personal opinions and does not necessarily reflect the views of the Deutsche Bundesbank or the Eurosystem.

[1] Theory is more ambiguous: it identifies factors which determine how much debt can or, in fact, should be sustained (e.g. Barro, 1979; Aiyagari and McGrattan, 1998; Mian, Straub and Sufi, 2025).

Panel A. Government debt per GDP



Panel B. Historical returns

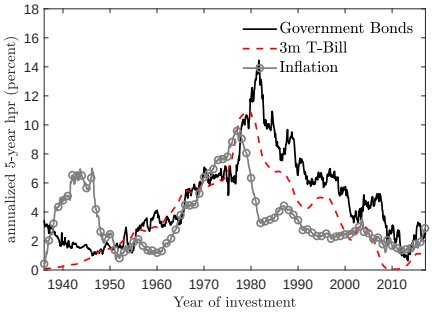


FIGURE 1. GOVERNMENT DEBT AND HISTORICAL RETURNS

Note: Left panel shows market value of interest bearing government debt in percent of GDP; right panel shows average annualized returns on different assets for a 5-year holding period: the ex-post holding period return of an average maturity government bond portfolio (solid black line) and the average annualized return of 3-month Treasury bills (dashed red line); the grey circle line shows average inflation over the same holding period. Sources: Hall et al. (2021) and FRED Economic Data.

In this paper, we ask if and, if so, how financial repression contributed to the evolution of the debt-to-GDP ratio in the U.S. after WW2. This is challenging because, first, the interest rate which would have prevailed in the absence of repression—say, the "laissez-faire" interest rate—is not directly observable. Second, financial repression not merely redistributes from bond holders to the government but impacts investment and savings decisions, too, a point stressed by Chari, Dovis and Kehoe (2020), or CDK for short. As a consequence, repression affects the evolution of the debt-to-GDP ratio not only directly by altering interest rates but also indirectly via growth, inflation, and tax revenues.

We build on CDK and model financial repression within a New Keynesian DSGE model that we estimate on U.S. time series data for the post-WW2 period. Our estimates show that repression was indeed pervasive but gradually phased out in the decades following the war. To understand its effects we construct a counterfactual, assuming that repression had not been phased out: In this case the debt-to-GDP ratio would have actually declined *less* because of lower GDP growth. Hence, if the point of repression was to bring down the debt-to-GDP ratio it was counterproductive. At the same time, we may rationalize earlier results suggesting that repression helped to bring down the debt-to-GDP ratio. In our framework, this holds, too, if we abstract from general equilibrium effects and consider the effect of financial repression on interest rates in isolation. In this case, we find it contributed to the decline of the debt-to-GDP ratio by around 26 percentage points, consistent with recent estimates by Acalin and Ball (2026).

Financial repression may come in various forms, as discussed in detail by Reinhart and Sbrancia (2015). One prominent example includes explicit caps on interest rates such as Regulation Q that limited banks' interest rate payments on

deposits, directing savings to bonds. Yet, the scope of financial repression is much broader as it aims at keeping long-term rates low. At some point this was explicit Fed policy. In the late 1940s, it allowed short rates to fluctuate, but maintained a ceiling of 2.5% for the long-term rate (Chandler, 1949). Similarly, during the early 1960s, "Operation Twist" sought to raise short-term rates while keeping long-term rates low. By end-1961, the Fed ended its "bills only" policy (Hetzel and Leach, 2001*b*,*a*). Moreover, to prevent arbitrage, financial repression requires a captive audience for government debt. In the case of banks, this may be the result of explicit regulatory requirements or implicitly via "moral suasion."[2]

We model financial repression as in CDK and assume a banking sector that faces a regulatory constraint: It mandates that banks hold a certain fraction of their assets as long-term government debt. In order to account for changes in the extent of financial repression, we allow this fraction to vary over time. This captures the notion that the banking sector is a captive audience for government debt while being agnostic about specific measures on which the government relies when it auctions off debt at elevated prices. CDK study conditions under which financial repression can be optimal. In particular, absent commitment, repression can enhance the credibility of the government as a borrower by raising the economic costs of default (see also Jeanne, 2023).

We abstract from default altogether and we embed financial repression à la CDK in a New Keynesian framework with financial intermediation, using Gertler and Karadi (2013) as our point of departure. Our model also features a rich fiscal sector along the lines of Bianchi and Ilut (2017). For these reasons it is able to capture well the joint dynamics of government debt, interest rates, and inflation—all of which are, in turn, key to understanding why real rates have been particularly low in the post-WW2 period. More generally, our framework allows us to shed light on the interaction between financial repression, on the one hand, and conventional monetary and fiscal policies, on the other hand. As a caveat, we note upfront that our model does not capture all aspects of financial repression. Instead, we focus on the impact of repression on banks' balance sheets in order to quantify its adverse impact on investment and economic activity. In this way, we not only highlight an aspect of financial repression that is typically ignored in the literature seeking to quantify its contribution to debt reduction, but also capture a key feature of financial repression: the willingness of the private sector to hold government debt at compressed yields.

We estimate the model on U.S. time series for the period 1948q2–1974q4 using standard macroeconomic and fiscal indicators but also time series specific to the banking sector: The share of outstanding government debt held by commercial banks and the share of government debt in banks' portfolios. We show that these series are key for identifying repression shocks. For our sample period, we find that the estimated model explains the data best under a fiscal-led regime

[2]As an example, Reinhart and Sbrancia (2015) relate an episode from the 1960s in which the Fed asked commercial banks—via mail—for foreign-credit restraint.

where monetary policy does not satisfy the Taylor principle, or put differently, fiscal policy is active and monetary policy is passive (Leeper, 1991). This finding confirms earlier results by, for example, Leeper, Traum and Walker (2017) and Bianchi and Ilut (2017). Based on the estimated model, we study in detail the mechanism through which repression affects the economy. By zooming in on the market for government debt, we first show how a repression shock—an exogenous tightening of the regulatory constraint—lowers the yield on government debt or, equivalently, raises demand for government debt at a given yield, thereby increasing fiscal space. Second, in general equilibrium, the effect of a repression shock on the debt-to-GDP ratio is ambiguous: it may go up or decline depending on parameter values. We establish this formally based on a prior-predictive analysis. For the estimated model, we find that a repression shock raises the debt-to-GDP ratio, as it crowds out investment and thereby reduces growth, offsetting the favorable effect of lower interest rates on the government budget. We emphasize, however, that this result is not hard-wired; it is driven by the data.

We quantify the contribution of repression shocks to the evolution of the debt-to-GDP ratio based on a historical decomposition. For this purpose, we use data from 1936 until the end of our estimation sample. This approach allows us to avoid estimating the model on World War II data while still including the period during which the build-up of debt was most pronounced in our analysis. We find that repression shocks contributed substantially to the increase in debt during the 1940s, accounting for roughly one quarter of the rise. Thereafter, negative repression shocks contributed to its decline, reflecting a gradual phase-out of repression. We also validate these results externally by linking the identified repression shocks to specific policy measures featured in the historical narrative on financial repression. In addition, we show that key model predictions align with data not used in the estimation, both unconditionally and conditional on the identified repression shocks.

To assess the impact of repression on the evolution of the debt-to-GDP ratio during our estimation period and the economy more broadly, we construct a counterfactual scenario based on the estimated model. We solve the model for the estimated parameters and for the estimated shocks but assume—counterfactually—that the regulatory constraint had not been relaxed. Instead we impose a constant value corresponding to the estimate for the beginning of our sample period. We solve the model, in other words, under the assumption that repression had not been phased out. Once we do that, we find that if repression had stayed high, the debt-to-GDP ratio would have declined *less*. In this scenario, at the end of 1974, investment and GDP would have been 12% and 5% lower, respectively. As a result, keeping repression high would not have contributed to the liquidation of government debt at all. Rather, we find that any direct effect of repression on interest rates is more than offset by general equilibrium effects. Notably, our counterfactuals take the fiscal position at the beginning of the sample as given. Our analysis therefore speaks to the question of what the overall impact of re-

pression was, *given* the debt overhang inherited in 1948.

Finally, to reconcile our analysis with existing results, we isolate the direct effect of repression on interest rates by computing the laissez-faire rate, the interest rate without financial repression. We find it considerably higher than actual interest rates. Also according to this measure, repression was phased out during our sample period: The gap between the laissez-faire and the actual rate was largest in the late 1940s and the early 1950s, declined afterwards and was closed by the late 1960s. Importantly, if we compute the counterfactual evolution of the debt-to-GDP ratio assuming the government had paid the laissez-faire rate, *all else equal*, we find the debt ratio would have declined much less than in the data. In other words, had the government paid the higher laissez-faire rate, the debt-to-GDP ratio would have declined by only 30 percentage points, rather than by the 57 percentage points which we observe during our sample period. From an accounting perspective it thus appears that repression was conducive to lowering the debt-to-GDP ratio—even though it was not.

The paper is structured as follows. In the remainder of the introduction we clarify the paper's connection to the existing literature. Section I presents our model which embeds financial repression à la CDK in a New Keynesian DSGE setup. Section II describes the data, the estimation as well as the estimation results. We inspect the mechanism in some detail in Section III, offer an external validation in Section IV, and present counterfactuals in Section V. A final section offers a short conclusion.

Related literature. First, we share the general-equilibrium perspective on repression in the recent literature, notably Chari, Dovis and Kehoe (2020), but focus on a purely positive analysis, whereas theirs is normative. They show that, without commitment, repression can be optimal when governments face exceptionally high fiscal needs, such as during wartime. Under commitment, however, the optimal level of repression—even in the presence of distortionary labor taxes—is zero, broadly in line with the positive results of our counterfactual analysis. Payne and Szőke (2025) study how the government may obtain a funding advantage through regulation (and/or repression), notably by creating a captive audience for government debt, as in our model. However, their framework relies on different microfoundations, as it is based on interbank settlement frictions. Moreover, it allows for the possibility that the monetary–fiscal policy mix erodes the government's funding advantage by undermining the hedging properties of government debt.

Second, in addition to the studies mentioned above, there is earlier work which analyzes repression in various contexts. Giovannini and de Melo (1993) consider developing economies and proxy the laissez-faire interest rate with the interest rate a government pays on world capital markets. They document that the "repression tax" contributed handsomely to government revenues. An earlier survey by Fry (1997) concludes that financial repression contributed to government rev-

enue in the order of 2 percent of GDP in a sample of developing economies. More recent empirical work suggests that repression has been under way during the recent euro area crisis (Becker and Ivashina, 2018; Ongena, Popov and Van Horen, 2019). Reinhart, Reinhart and Rogoff (2015) study debt reduction policies in 22 advanced economies from 1800 to 2014 and find that governments have often relied on "heterodox approaches". Yet all these studies take an "all-else-equal" perspective on financial repression, while we account for its general-equilibrium effects.

Third, there are alternative ways to think of financial repression. In an early study, Roubini and Sala-i-Martin (1995) put forward a model where financial repression raises money demand, say because of regulation that limits the use of checks, ATMs etc. This in turn raises the base on which the inflation tax operates. As a result, their model predicts that inflation and financial repression go hand in hand, quite contrary to what our analysis suggests (see also Brock, 1989). More generally, financial regulation has been found to impact financial markets. Du, Tepper and Verdelhan (2018), for instance, rationalize large and persistent deviations from covered interest rate parity in light of the new regulatory environment put in place after the global financial crisis. It impairs the ability of financial intermediaries to arbitrage away spreads between the return of riskless securities, a mechanism that operates at the heart of our model, too.

Fourth, while repression lowers the return on government debt, other factors may do so as well. In particular, government debt may be perceived as especially safe and liquid and thus carry a convenience yield, which has been extensively studied in the literature (Krishnamurthy and Vissing-Jorgensen, 2012; Jiang, Krishnamurthy and Lustig, 2023; Lehner et al., 2025). While this suggests that financial repression and alternative mechanisms may be observationally equivalent in the data, we show that the repression shocks we identify are closely linked to the historical narrative on repression. More broadly, there is a large literature on the drivers of real interest rates, particularly in recent decades (see, e.g., Del Negro et al., 2017). In addition, Ilut, Krivenko and Schneider (2024) emphasize the interplay between financial frictions and uncertainty in explaining persistently low real interest rates. As a caveat, our linearized rational expectations framework does not allow us to capture the effects of uncertainty on saving behavior.

I. The model

Our analysis relies on a medium-scale New Keynesian model in which the financial and fiscal sectors take center stage. In modelling these sectors we build on earlier work by Gertler and Karadi (2013), Bianchi and Ilut (2017), and Chari, Dovis and Kehoe (2020). The economy is populated by four types of agents: households, banks, firms, and a government sector. We discuss their decision problems in some detail in what follows.

A. Households

There is a continuum of identical households which consume, save and supply labor to an employment agency. As in Gertler and Karadi (2011, 2013), a fraction f of household members are bankers and a fraction $1-f$ are workers. Workers are employed by an intermediate good firm and earn wage income. Bankers manage a financial intermediary, which collects deposits from all households, funds non-financial firms, and holds government bonds. There is perfect consumption smoothing within the household. Over time, each member may change its occupation, yet the fraction of household members in each occupation remains constant. In particular, with probability $1-\sigma$ a banker quits and becomes a worker next period, while with probability $f(1-\sigma)$ a worker becomes a banker. Once the banker exits its business, retained earnings are transferred to the household and the bank shuts down. This setup ensures that financial intermediaries are unable to finance all investment projects with retained earnings and thus remain dependent on deposits. Any new banker obtains constant startup funds, X, from the household. The representative household maximizes expected lifetime utility,

$$E_t \sum_{t=0}^{\infty} \beta^t e^{z_{d,t}} \left[\ln\left(c_t - b c_{t-1}^a\right) - \Psi_L \frac{l_t^{1+\psi}}{1+\psi} \right], \tag{1}$$

with consumption, c_t, the predetermined stock of consumption habits, c_{t-1}^a, labor l_t, and preference parameters ψ and Ψ_L. The habit stock is external to the household; thus c_t^a denotes the average consumption in the economy, with parameter $b \in (0,1)$ determining the extent of external habits. Parameter $\beta \in (0,1)$ is the discount factor and $z_{d,t}$ a preference shock that follows an AR(1) process:

$$z_{d,t} = \rho_d z_{d,t-1} + \sigma_d \epsilon_{d,t}, \text{ with } \epsilon_{d,t} \overset{iid}{\sim} N(0,1). \tag{2}$$

The household budget constraint reads in nominal terms as follows:

$$\begin{aligned} P_t w_t l_t + Div_t - P_t X - P_t T_t + P_t TR_t + R_{t-1}^d D_{t-1} + R_t^b Q_{t-1}^b B_{t-1}^h + R_t^k Q_{t-1}^s S_{t-1}^h \\ \geq P_t c_t + D_t + Q_t^b \left(B_t^h + f_b\left(B_t^h\right)\right) + Q_t^s \left(S_t^h + f_k\left(S_t^h\right)\right). \end{aligned} \tag{3}$$

The aggregate price level is given by P_t. The household earns labor income, given by hours worked, l_t, and the real wage, w_t. It receives payouts from ownership of non-financial firms and financial firms, Div_t, pays taxes, T_t, and receives transfers, TR_t, from the government.

The household earns income on three kinds of assets: deposits, D_t, which earn a predetermined risk-free nominal return R_{t-1}^d; government bonds, B_t^h, and private securities, S_t^h, which trade at prices Q_t^b and Q_t^s, respectively. Government debt is a perpetuity with geometrically decaying coupons as in Woodford (2001) in

order to capture the maturity of government debt. The realized nominal return on government debt with an average maturity is given by

$$R_t^b = \frac{1 + \iota Q_t^b}{Q_{t-1}^b}, \tag{4}$$

where $\iota \in (0, 1)$ denotes the rate of decay. Regarding the household's private security holdings, we let r_t^k denote the rental rate of capital which is the net period income flow from financing a unit of capital (see below for further details). Then, the realized nominal return of the private security, R_t^k, is given by

$$R_t^k = \frac{r_t^k + (1 - \delta) Q_t^s}{Q_{t-1}^s} \Pi_t, \tag{5}$$

where Q_t^s is the market value of the security, δ the depreciation rate of a unit of capital, and $\Pi_t = P_t / P_{t-1}$ inflation.

Adjusting the portfolio is costly for the household. Specifically, as in Coenen et al. (2018) we assume:

$$f_k\left(S_t^h\right) = 1/2\kappa_k \left(S_t^h - \tilde{S}^h\right)^2 / \tilde{S}^h. \tag{6}$$

$$f_b\left(B_t^h\right) = 1/2\kappa_b \left(B_t^h - \tilde{B}^h\right)^2 / \tilde{B}^h, \tag{7}$$

where parameters $\tilde{S}^h$ and $\tilde{B}^h$ determine to which amount asset holding is costless and κ_k and κ_b govern the degree of sluggishness in portfolio rebalancing. As emphasized by Coenen et al. (2018), such costs are essential in medium-scale DSGE models with financial intermediaries. Without them, households would arbitrage away any spread between returns on government debt and private securities each period, eliminating the wedge created by financial frictions in the banking sector. The adjustment costs therefore generate limited asset-market arbitrage.

Optimality requires consumption, labor supply, and deposits to satisfy the following:

$$\Lambda_t = e^{z_{d,t}} \left(c_t - b c_{t-1}^a\right)^{-1} \tag{8}$$

$$w_t = e^{z_{d,t}} \Psi_L \frac{l_t^\psi}{\Lambda_t} \tag{9}$$

$$1 = R_t^d E_t \left[M_{t,t+1} \Pi_{t+1}^{-1}\right], \tag{10}$$

where $M_{t,t+1} = \beta \Lambda_{t+1} / \Lambda_t$ is the household's real stochastic discount factor and

Λ_t the marginal utility of consumption. The optimal portfolio choice satisfies:

$$S_t^h = \left(1 + E_t\left[\frac{M_{t,t+1}\left(R_{t+1}^k - R_t^d\right)}{\Pi_{t+1}\kappa_k}\right]\right)\tilde{S}^h \tag{11}$$

$$B_t^h = \left(1 + E_t\left[\frac{M_{t,t+1}\left(R_{t+1}^b - R_t^d\right)}{\Pi_{t+1}\kappa_b}\right]\right)\tilde{B}^h + z_{b,t}. \tag{12}$$

The first order condition of households' bond holdings features a Treasury-demand shock, $z_{b,t}$, which evolves according to

$$z_{b,t} = \rho_b z_{b,t-1} + \sigma_b \epsilon_{b,t}, \text{ with } \epsilon_{b,t} \overset{iid}{\sim} N(0,1). \tag{13}$$

It captures time-variation in households' appetite for Treasuries arising from factors not explicitly modeled—such as changes in perceived safety or liquidity premia—and enables the model to account for fluctuations in the composition of government bond holdings.

B. Financial intermediaries

A representative financial intermediary, or bank, i relies on real deposits, $d_{i,t} = D_{i,t}/P_t$, and retained earnings to fund either the capital stock of non-financial firms or holdings of government debt. Letting $s_{i,t}^b$ denote the funding of non-financial firms by bank i, $b_{i,t}^b$ the holdings of government debt and $n_{i,t}$ its real equity position (or real net worth), the balance sheet of the intermediary is given by:

$$Q_t^s s_{i,t}^b + Q_t^b b_{i,t}^b = d_{i,t} + n_{i,t}, \tag{14}$$

where $s_{i,t}^b = S_{i,t}^b/P_t$ and $b_{i,t}^b = B_{i,t}^b/P_t$. Net worth evolves as follows:

$$n_{i,t+1} = \frac{1}{\Pi_{t+1}}\left[\left(R_{t+1}^k - R_t^d\right)Q_t^s s_{i,t}^b + \left(R_{t+1}^b - R_t^d\right)Q_t^b b_{i,t}^b + R_t^d n_{i,t}\right]. \tag{15}$$

The bank maximizes the expected value of its terminal value net worth:

$$V_{i,t} = \max\left\{(1-\sigma)E_t\sum_{j=1}^{\infty}\sigma^{j-1}M_{t,t+j}n_{i,t+j}\right\}, \tag{16}$$

where $1-\sigma$ is the exogenous probability that a banker will close down its operations. Maximization is subject to two constraints. The first reflects the agency friction introduced by Gertler and Karadi (2011). At any point in time, a banker

may divert a fraction $\theta \in (0,1)$ of private-sector funding and a fraction $\Delta\theta$ of its debt holdings, where $\Delta \in (0,1)$.[3] Then, for depositors to be willing to lend to the bank, we require the following *participation constraint* to be satisfied:

$$V_{i,t} \geq \theta \left(Q_t^s s_{i,t}^b + \Delta Q_t^b b_{i,t}^b \right). \tag{17}$$

Following Chari, Dovis and Kehoe (2020) we assume an additional *regulatory constraint*:

$$Q_t^b b_{i,t}^b \geq \tilde{\Gamma}_t \left(Q_t^b b_{i,t}^b + Q_t^s s_{i,t}^b \right). \tag{18}$$

Here, $\tilde{\Gamma}_t$ is the minimum share of assets that banks need to hold as government debt. It captures a variety of measures, such as those discussed by Reinhart and Sbrancia (2015). These measures may not explicitly require banks to hold a certain fraction of assets in government debt but effectively raise the demand for—and thus the price of—government debt, which is precisely the implication of the regulatory constraint. We rearrange it slightly to

$$Q_t^b b_{i,t}^b \geq \Gamma_t Q_t^s s_{i,t}^b, \tag{19}$$

with $\tilde{\Gamma}_t = \Gamma_t / (1 + \Gamma_t)$. The regulatory constraint adjusts exogenously due to *repression shocks*:

$$\Gamma_t = (1 - \rho_\Gamma)\, \bar{\Gamma} + \rho_\Gamma \Gamma_{t-1} + \sigma_\Gamma \epsilon_{\Gamma,t}, \text{ with } \epsilon_{\Gamma,t} \overset{iid}{\sim} N(0,1). \tag{20}$$

Optimality requires:

$$E_t M_{t,t+1} \left(R_{t+1}^k - R_t^d \right) \frac{\Omega_{i,t+1}}{\Pi_{t+1}} = \frac{\lambda_{i,t}\theta + \mu_{i,t}\Gamma_t}{1 + \lambda_{i,t}} \tag{21}$$

$$E_t M_{t,t+1} \left(R_{t+1}^b - R_t^d \right) \frac{\Omega_{i,t+1}}{\Pi_{t+1}} = \frac{\lambda_{i,t}\theta\Delta - \mu_{i,t}}{1 + \lambda_{i,t}}, \tag{22}$$

where $\Omega_{i,t+1} = 1 - \sigma + \sigma(\partial V_{i,t+1}/\partial n_{i,t+1})$ is the shadow value of one unit of net worth to the bank; $\lambda_{i,t}$ and $\mu_{i,t}$ are the multipliers on the participation and on the regulatory constraint, respectively. As such they measure the extent to which the two financial frictions constrain banks' behavior.

Against this background, a few remarks are in order. In particular, equation (21) relates the (expected) excess return on investment in intermediate-good firms (over the deposit rate) to the tightness of the participation constraint (17) and

[3] This assumption reflects the notion that private-sector funding is easier to divert because it is harder for depositors to observe.

regulatory constraint (19). Intuitively, to the extent that bankers are leverage-constrained, expected excess returns persist in equilibrium. Additionally, by distorting banks' portfolio choices through government regulation, a binding regulatory constraint (i.e., $\mu_{i,t} > 0$) reduces demand for real capital, thereby raising excess returns. This leads to a crowding out of investment and reduces real activity as pointed out by Chari, Dovis and Kehoe (2020).

Equation (22), in turn, relates the (expected) excess return on government debt to the tightness of both constraints. In our setup, there is market segmentation because households find it costly to adjust their debt holdings and banks are leverage-constrained. As a result, there are limits to arbitrage and differences in expected yields persist in equilibrium. In addition there is a "regulatory discount" which appears in equation (22) via $\mu_{i,t}$. Recall that this is the multiplier on the regulatory constraint. The tighter the constraint (19), the lower the expected excess return on government debt. Intuitively, to the extent that the regulatory constraint binds, the price of government debt is pushed up and (expected) returns are depressed—in other words, financial repression.

The limits to arbitrage stem from the restrictions that the participation and regulatory constraints place on the size of a bank's portfolio relative to its net worth, the bank's "risk-weighted" leverage ϕ_t:[4]

$$\phi_t = \frac{Q_t^s s_{i,t}^b + \Delta Q_t^b b_{i,t}^b}{n_{i,t}}, \tag{23}$$

$$\text{with } \phi_t = \frac{E_t M_{t,t+1} \Omega_{i,t+1} \Pi_{t+1}^{-1} R_t^d}{\theta - E_t \frac{M_{t,t+1}\Omega_{i,t+1}}{(1+\Delta\Gamma_t)\Pi_{t+1}} \left[R_{t+1}^k - R_t^d + \Gamma_t \left(R_{t+1}^b - R_t^d\right)\right]}. \tag{24}$$

C. Firms

Firms produce a final good under perfect competition. It is an aggregate of intermediate goods, indexed by j:

$$y_t = \left(\int_0^1 y_{j,t}^{\frac{\theta_p - 1}{\theta_p}} dj \right)^{\frac{\theta_p}{\theta_p - 1}}, \tag{25}$$

where $\theta_p > 1$ is the elasticity of substitution across the intermediate goods. A representative firm takes the price of output, P_t, and the price of inputs, $P_{j,t}$, as given. The resulting demand function for the intermediate good is

$$y_{j,t} = \left(\frac{P_{j,t}}{P_t} \right)^{-\theta_p} y_t, \tag{26}$$

[4]The online appendix provides more detailed derivations.

and the aggregate price level is defined as

$$P_t = \left(\int_0^1 P_{j,t}^{1-\theta_p} dj \right)^{\frac{1}{1-\theta_p}}. \tag{27}$$

Intermediate goods are produced by firms which operate under monopolistic competition. The production function for a generic good j is given by:

$$y_{j,t} = k_{j,t-1}^{\alpha} \left(A_t l_{j,t} \right)^{1-\alpha} - A_t \Phi, \tag{28}$$

where $k_{j,t}$ and $l_{j,t}$ denote capital services and labor, respectively. $\alpha \in (0,1)$ and $\Phi > 0$. The latter parameterizes the fixed costs of production that grow at the rate of technology progress. The variable A_t represents productivity and follows a stochastic trend. Specifically, defining $\exp\{z_{a,t}\} = A_t / A_{t-1}$, we assume

$$z_{a,t} = (1-\rho_a)\, \bar{z}_a + \rho_a z_{a,t-1} + \sigma_a \epsilon_{a,t}, \text{ with } \epsilon_{a,t} \overset{iid}{\sim} N(0,1). \tag{29}$$

Firms face perfectly competitive factor markets for capital and labor. Cost minimization implies that firms have identical marginal costs:

$$mc_t = \left(\frac{1}{1-\alpha} \right)^{1-\alpha} \left(\frac{1}{\alpha} \right)^{\alpha} \left(\frac{w_t}{A_t} \right)^{1-\alpha} \left(r_t^k \right)^{\alpha}, \tag{30}$$

$$\frac{k_{t-1}}{l_t} = \frac{w_t}{r_t^k} \frac{\alpha}{(1-\alpha)}, \tag{31}$$

where r_t^k and w_t represent the rental rate of capital and real wages, respectively, and mc_t denotes real marginal costs. The law of motion for capital is given by

$$k_t = (1-\delta)\, k_{t-1} + \mathcal{I}_t, \tag{32}$$

where δ is the depreciation rate and $\mathcal{I}_t$ new capital. This is produced by investment goods producers based on the following production function:

$$\mathcal{I}_t = e^{z_{I,t}} \left[1 - S\left(\frac{I_t}{I_{t-1}} \right) \right] I_t, \tag{33}$$

where $S(\cdot)$ is an investment adjustment costs function as in Christiano, Eichenbaum and Evans (2005). We assume that $S(\cdot) = S'(\cdot) = 0$ and $\nu \equiv S''(\cdot) > 0$ along the steady state growth path with the adjustment cost parameter ν. Investment costs are subject to an investment-specific efficiency shock $z_{I,t}$ which

evolves according to the process:

$$z_{I,t} = \rho_I z_{I,t-1} + \sigma_i \epsilon_{I,t}, \text{ with } \epsilon_{I,t} \overset{iid}{\sim} N(0,1). \tag{34}$$

Price setting is constrained à la Calvo (1983). In a given period, firms are allowed to adjust prices with probability $(1-\gamma_p)$ only. If a firm cannot adjust, its price evolves according to the indexation rule: $P_{j,t} = P_{j,t-1}\Pi_{t-1}^{\xi_p}$. If the firm is able to adjust its price, the price $\tilde{p}_t = P_{j,t}$ is set to maximize the value of its expected future dividend stream subject to the demand for its products. Optimality requires

$$\mathcal{K}_t^p = y_t \tilde{p}_t + \gamma_p E_t \left[M_{t,t+1} \left(\frac{\Pi_t^{\xi_p}}{\Pi_{t+1}} \right)^{1-\theta_p} \frac{\tilde{p}_t}{\tilde{p}_{t+1}} \mathcal{K}_{t+1}^p \right] \tag{35}$$

and

$$\mathcal{F}_t^p = y_t mc_t + \gamma_p E_t \left[M_{t,t+1} \left(\frac{\Pi_t^{\xi_p}}{\Pi_{t+1}} \right)^{-\theta_p} \mathcal{F}_{t+1}^p \right] \tag{36}$$

with

$$\mathcal{F}_t^p = \frac{\theta_p - 1}{\theta_p} \mathcal{K}_t^p, \tag{37}$$

which is the same for all firms that can adjust their price in period t. The evolution of the aggregate price index is implicitly given by:

$$1 = \gamma_p \left(\frac{\Pi_{t-1}^{\xi_p}}{\Pi_t} \right)^{1-\theta_p} + (1-\gamma_p) (\tilde{p}_t)^{1-\theta_p}. \tag{38}$$

D. Government

In each period the government finances purchases g_t, transfers TR_t and interest rate payments by raising taxes T_t and issuing nominal government debt B_t. Using $\mathcal{B}_t \equiv Q_t^b B_t / (P_t y_t)$ to denote the debt-to-GDP ratio, e_t the expenditure ratio (sum of purchases and transfers relative to GDP) and τ_t taxes relative to GDP, we write the period budget constraint of the government as follows:

$$\mathcal{B}_t + \tau_t = R_t^b \frac{\mathcal{B}_{t-1} y_{t-1}}{\Pi_t y_t} + e_t + z_{tp,t}. \tag{39}$$

Following Bianchi and Ilut (2017), the budget constraint includes a disturbance $z_{tp,t}$ that captures factors not explicitly modeled, such as shifts in the maturity structure or movements in the term premium. It evolves according to:

$$z_{tp,t} = \rho_{tp} z_{tp,t-1} + \sigma_{tp} \epsilon_{tp,t}, \text{ with } \epsilon_{tp,t} \overset{iid}{\sim} N(0,1). \tag{40}$$

We also follow Bianchi and Ilut (2017) as we specify the process for expenditures and purchases. Expenditures consist of a short-term component $\tilde{e}_t^S$ and a long-term component $\tilde{e}_t^L$ (i.e., $\tilde{e}_t = \tilde{e}_t^L + \tilde{e}_t^S$):[5]

$$\tilde{e}_t^S = \rho_{eS} \tilde{e}_{t-1}^S + (1-\rho_{eS}) \eta_{eS} \hat{y}_t + \sigma_{eS} \epsilon_{eS,t}, \text{ with } \epsilon_{eS,t} \overset{iid}{\sim} N(0,1) \tag{41}$$

$$\tilde{e}_t^L = \rho_{eL} \tilde{e}_{t-1}^L + \sigma_{eL} \epsilon_{eL,t}, \text{ with } \epsilon_{eL,t} \overset{iid}{\sim} N(0,1). \tag{42}$$

The short-term component captures cyclical variation, while the long-term component follows a highly persistent AR(1) process capturing large, long-lasting government programs not explicitly modeled (e.g., the "Great Society" and war spending). Government purchases relative to expenditures, defined as $\chi_t = g_t / E_t$, evolve according to:

$$\tilde{\chi}_t = \rho_\chi \tilde{\chi}_{t-1} + (1-\rho_\chi) \eta_{\chi,y} \hat{y}_t + \sigma_\chi \epsilon_{\chi,t}, \text{ with } \epsilon_{\chi,t} \overset{iid}{\sim} N(0,1) \tag{43}$$

which allows for a contemporaneous feedback effect from output ($\eta_{\chi,y}$).

For monetary policy we assume a conventional interest-rate feedback rule:

$$R_t = \rho_R R_{t-1} + (1-\rho_R)\left(\bar{R} + \eta_y \hat{\Xi}_t + \eta_\pi \hat{\Pi}_t\right) + \sigma_m \epsilon_{m,t}, \text{ with } \epsilon_{m,t} \overset{iid}{\sim} N(0,1), \tag{44}$$

where ρ_R is a positive constant and η_π and η_y capture the reaction coefficients to inflation and output growth ($\Xi_t = y_t / y_{t-1}$), respectively. Finally, taxes adjust as follows:

$$\tilde{\tau}_t = \rho_\tau \tilde{\tau}_{t-1} + (1-\rho_\tau)\left[\eta_b \tilde{\mathcal{B}}_{t-1} + \eta_{\tau,e} \tilde{e}_t + \eta_{\tau,y} \hat{y}_t\right] + \sigma_\tau \epsilon_{\tau,t}, \text{ with } \epsilon_{\tau,t} \overset{iid}{\sim} N(0,1), \tag{45}$$

where $\tilde{\tau}_t$ are tax revenues, measured in terms of GDP. There is adjustment to expenditures, output, and debt, according to coefficients $\eta_{\tau,e}$, $\eta_{\tau,y}$ and η_b. The last coefficient together with the response coefficient in the interest-rate rule η_π determines the specific monetary-fiscal regime, an issue we take up in Section II.B below.

[5] Throughout, $\hat{x}_t = \log(x_t/\bar{x})$ denotes percentage deviations from steady state, and $\tilde{x}_t = x_t - \bar{x}$ linear deviations.

E. Aggregation and market clearing

For market clearing, we assume that deposits earn the risk-free rate set by monetary policy: $R_t^d = \exp(R_t)$. Moreover, the total amount of private securities and long-term government bonds outstanding are given by

$$s_t = s_t^h + s_t^b, \tag{46}$$

$$b_t = b_t^h + b_t^b. \tag{47}$$

Additionally, for market clearing in markets for private securities, we assume that the total supply of private securities s_t is equal to the capital stock k_t. Because the bank's leverage ϕ_t does not depend on individual bank-level factors, we can aggregate the bank's balance sheet conditions into an aggregate balance sheet condition:

$$\phi_t = \frac{Q_t^s s_t^b + \Delta Q_t^b b_t^b}{n_t}. \tag{48}$$

Similarly, since the bank's real net worth is linear in the bank's choice variables, we can simply aggregate it. Because those banks that close down are randomly chosen, the sum of surviving banks is simply proportional to the aggregate. Newly born intermediaries are given, in aggregate, X of real start-up net worth.

$$\begin{aligned} n_t = \frac{\sigma}{\Pi_t} \Big[& \left(R_t^k - R_{t-1}^d \right) Q_{t-1}^s s_{t-1}^b \\ & + \left(R_t^b - R_{t-1}^d \right) Q_{t-1}^b b_{t-1}^b + R_{t-1}^d n_{t-1} \Big] + A_t X. \end{aligned} \tag{49}$$

Good market clearing at the level of intermediates implies

$$p_t^+ y_t = k_{t-1}^\alpha \left(A_t l_t \right)^{1-\alpha} - A_t \Phi, \tag{50}$$

where $l_t = \int_0^1 l(j,t)\ dj$ and $k_t = \int_0^1 k_{j,t}\ dj$ are the aggregate labor and capital inputs, respectively. The term $p_t^+ = \int_0^1 \left(P_{j,t}/P_t \right)^{-\theta_p}\ dj$ measures the price dispersion arising from staggered price setting. Price dispersion follows the law of motion

$$p_t^+ = (1-\gamma_p) \left(\tilde{p}_t \right)^{-\theta_p} + \gamma_p \left(\frac{\Pi_{t-1}^{\xi_p}}{\Pi_t} \right)^{-\theta_p} p_{t-1}^+. \tag{51}$$

Finally, at the aggregate level, the following resource constraint needs to be sat-

isfied

$$y_t = c_t + I_t + g_t. \tag{52}$$

II. Estimation

We estimate the model using Bayesian techniques with data from 1948Q2 to 1974Q4, a period characterized by a sizable reduction in government debt. For years prior to 1948, the model is less well suited because the wartime economy potentially differs substantially from the peacetime environment it is designed to capture, and comprehensive high-quality data are lacking. Nevertheless, we use longer time series beginning in 1936 to perform a historical decomposition and assess the contribution of financial repression to the evolution of the debt-to-GDP ratio, whose build-up was particularly pronounced during the war. We next describe the dataset, outline the priors, and report the estimation results.

A. Data

We use ten quarterly time series to estimate the model: four macroeconomic indicators, four fiscal indicators, and two series capturing developments in the banking sector. Specifically, the macroeconomic indicators are real GDP growth, real investment growth, the annualized GDP deflator, and the short-term nominal interest rate. Since our fiscal sector closely follows Bianchi and Ilut (2017), our choice of fiscal indicators mirrors theirs. In particular, the estimation includes the annualized debt-to-GDP ratio, federal tax revenues and federal spending (both relative to GDP), and a broad measure of government expenditures, also relative to GDP and discussed in more detail below. Together, these series identify the fiscal block of the model.

To identify the financial block—particularly the repression shock and the Treasury-demand shock—we rely on two time series: commercial banks' holdings of government debt as a share of banks' total assets and as a share of outstanding public debt. Figure 2 shows both series. It illustrates the importance of government debt for banks' balance sheets (Panel A) and the importance of banks in the market for government debt (Panel B). These series allow us to identify the financial shocks because a repression shock forces banks to tilt their portfolios toward government debt, thereby raising their debt-to-asset ratio, whereas a Treasury-demand shock is primarily absorbed by households, altering the relative importance of banks in the government debt market.[6]

[6]See online appendix, which reports impulse response functions at the posterior mode of the estimated model for a repression shock and a Treasury-demand shock; we also show the responses to a negative household time-preference shock. The figure shows that bank holdings of government debt, measured relative to bank assets and as a fraction of total debt, distinguish the repression shock from the Treasury-demand shock, while changes in the level of debt holdings by households and banks distinguish the preference shock from both financial shocks.

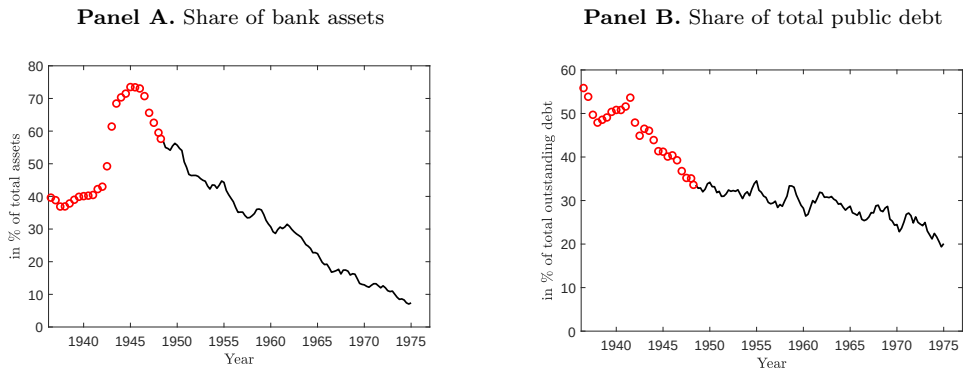


FIGURE 2. COMMERCIAL BANKS' HOLDINGS OF GOVERNMENT DEBT

Note: Figure shows U.S. government obligations as fraction of bank assets (total loans and investments held by all commercial banks) in the left panel and as fraction of the par value of government interest-bearing debt held by private investors from Hall et al. (2021) in the right panel (both series from the Board of Governors of the Federal Reserve System, provided by the FRASER Digital Library). Solid line: estimation sample. Circles: sample extended to perform the historical decomposition.

B. Calibration and prior choice

Prior to estimation, we set parameters that are not central to our analysis to customary values which we summarize in Table 1. A period in the model represents one quarter. The top panel reports parameters governing private-sector behavior. The discount factor β is set to 0.9985, the capital income share α to 1/3, and the depreciation rate of private capital δ to 0.025, implying an annual depreciation rate of 10 percent. The price markup is fixed at 10 percent, steady-state labor supply is calibrated to 1/3, and the Frisch elasticity of labor supply is set to 1.

In the second panel of the table, we report the steady-state values of the fiscal variables. They are set equal to their mean values during our sample period. The steady-state value of debt relative to (quarterly) GDP is set to 1.47. Federal government purchases as a share of GDP, g/y, and federal expenditures, e/y, are set to 0.11 and 0.165, respectively. We set the parameter capturing the persistence of the long-term component of government expenditure, ρ_{eL}, to 0.95 and the corresponding standard deviation of its innovations to $\sigma_{eL} = 0.1$, similar to Bianchi and Ilut (2017).

In the bottom panel of Table 1, we report the values for parameters which capture key aspects of household behavior and bank portfolios. We set the parameter ι so that the average maturity of government debt is 20 quarters, which corresponds to the U.S. average between 1952 and 2020 as reported by the Treasury Bulletin, see also Leeper, Traum and Walker (2017) or Bianchi and Ilut (2017).[7]

[7]Hall and Sargent (2011) and Acalin and Ball (2026) discuss changes in the average maturity of outstanding government debt. In particular, they find that the average maturity increased in the after-

We set the portfolio adjustment cost of private capital, κ_k, to 0.1 and estimate the corresponding portfolio adjustment cost for government debt. Given these parameters, the constants $\tilde{B}^h$ and $\tilde{S}^h$ in the portfolio adjustment cost functions of the households are chosen such that, in steady state, 70 percent of total government bonds and 85 percent of total private securities are held by households. This implies that commercial banks hold 30 percent of government debt in steady state, consistent with data for the estimation period, see Panel B of Figure 2.

We set the ratio of net worth to total assets to 10 percent, corresponding to the historical average for commercial banks over our sample period. The annualized spread on corporate loans and the spread on government bonds are set to 200 basis points and 50 basis points, respectively. For corporate loans, this is in line with Moody's Seasoned BAA corporate bond spreads between 1947 and 1974. For government bonds, the value is slightly smaller than the spread between the five-year government bond yield and the three-month T-bill, which averaged 0.6 percentage points between 1962 and 1974. The survival rate of bankers, σ, is set to 0.95, as in Sims and Wu (2021). We set Δ, which captures the proportional advantage in the seizure rate of government debt, to 0.5, as in Gertler and Karadi (2013).

This calibration implies values for the parameters X, Γ, μ, and θ. In particular, for θ, which denotes the fraction of capital that can be diverted, we obtain a value of 0.54, which lies between the values used by Gertler and Karadi (2013) and Sims and Wu (2021). The new equity infusion to intermediaries, X, amounts to about 0.93 percent of total equity. Γ is equal to 0.31, which implies that banks' holdings of government bonds amount to around 23.5 percent of total assets, $\Gamma/(\Gamma+1)$, in line with the average over our estimation period (see Panel A of Figure 2). Finally, in steady state financial repression lowers the annual return on government bonds by roughly 43 basis points.

We estimate the remaining parameters. For this purpose we set prior distributions in line with the literature. In what follows, we highlight those that are particularly relevant. Following Leeper, Traum and Walker (2017), we focus on the two distinct regions of the parameter subspace which yield unique bounded rational expectations equilibria. Applying the terminology by Leeper (1991), the regions are characterized by either an active monetary/passive fiscal policy regime (Regime M) or by a passive monetary/active fiscal policy regime (Regime F). To this end, we consider two sets of priors for the policy parameters which place nearly all probability mass on regions of the parameter space consistent with either Regime F or with Regime M. In particular, the prior for the Taylor-rule parameter η_π is beta-distributed with mean 0.5 under Regime F and gamma-distributed with mean 1.8 under Regime M, respectively. At the same time, the response coefficient in the tax rule, $\eta_{\tau,b}$, is set to 0 under Regime F and assumed

math of World War II before the trend reversed following the Treasury–Fed Accord of March 1951 and continued to decline until the mid-1970s. The Treasury Bulletin reports an average maturity of about 6.5 years in 1951 and about 2.9 years in 1974.

Table 1—Calibrated parameters.

Name	Symbol	Value
Discount factor	β	0.9985
Capital share in production	α	1/3
Depreciation rate	δ	0.025
Price markup	$\theta_p/\left(\theta_p-1\right)$	1.10
Steady-state labor supply	$\bar{l}$	1/3
Frisch elasticity of labor supply	$1/\psi$	1
Steady-state government debt (annualized)	$\bar{b}/4\bar{y}$	0.375
Steady-state government purchases	$\bar{g}/\bar{y}$	0.11
Steady-state government expenditures	$\bar{e}/\bar{y}$	0.165
AR parameter long-run exp.	ρ_{eL}	0.95
S.d. long-run expenditures	$100\sigma_{eL}$	0.1
Duration government debt (in quarters)	$(1-\iota\beta)^{-1}$	20
Portfolio adjustment cost capital	κ_k	0.1
Share of government bond holdings by households	$\bar{B}^h/\bar{B}$	0.70
Share of private capital holdings by households	$\bar{S}^h/\bar{S}$	0.85
Steady-state net worth/total assets ratio	$n/\left(Q^s k^b + Q^b b^b\right)$	0.1
Spread nominal return capital/deposit rate	$400(R^k - R^d)$	2
Spread nominal government debt/deposit rate	$400(R^b - R^d)$	0.5
Surviving probability of a banker	σ	0.95
Proportional advantage in seizure rate of government debt	Δ	0.5

to be gamma-distributed with mean 0.1 under Regime M.

We also highlight the prior choice for the parameter that captures the costs of adjusting households' holdings of government bonds. These adjustment costs are a crucial determinant of the overall level of financial frictions and, in particular, of financial repression. For example, if adjustment costs are zero, households can flexibly purchase government bonds from banks, which relaxes banks' balance-sheet constraints. We assume that κ_b follows a gamma distribution with mean 0.01 and standard deviation 0.075, in line with, but less restrictive than, the prior used in Coenen et al. (2018). The right panel of Table 2 summarizes these and the remaining priors for Regime F, which we find to be preferred by the data.

C. Estimation results

We approximate the posterior distribution using a random-walk Metropolis –Hastings algorithm. Specifically, we run two chains of 400k draws each and discard the first 80% as burn-in. Table 2 reports the posterior mode and 90 percent credible intervals for the estimated parameters under Regime F. In what follows, we focus much of our analysis on this regime, as it is somewhat favored by the data.[8] Bianchi and Ilut (2017), in related work, find more conclusive evidence

[8]We compute the log marginal data densities using the modified harmonic mean estimator of Geweke (1999). The densities for Regime F and Regime M are 3181.04 and 3178.46, respectively. This corresponds

Table 2—Prior and posterior distribution of parameters (Regime F)

		Posterior				Prior		
Name	Symbol	Mode	Mean	5%	95%	Type	Mean	Std
Calvo parameter	γ_p	0.544	0.545	0.489	0.600	B	0.5	0.05
Price indexation	ζ_p	0.272	0.312	0.162	0.460	B	0.3	0.10
Investment adjustment	ν	4.684	4.783	3.628	5.999	G	4.0	0.75
Habit formation	b	0.446	0.451	0.370	0.533	B	0.75	0.10
Portfolio adj. cost	κ_b	0.033	0.038	0.019	0.056	G	0.01	0.0075
Interest rate AR coefficient	ρ_R	0.828	0.830	0.797	0.865	B	0.8	0.1
Interest rate inflation coefficient	η_π	0.719	0.701	0.605	0.797	B	0.5	0.1
Interest rate output coefficient	η_y	0.030	0.047	0.005	0.087	G	0.15	0.1
AR coefficient Tax rate	ρ_τ	0.678	0.679	0.615	0.741	B	0.5	0.1
Tax rate expenditure coefficient	$\eta_{\tau,e}$	0.216	0.227	0.089	0.352	N	0.5	0.2
Tax rate output coefficient	$\eta_{\tau,y}$	0.115	0.118	0.036	0.194	N	0.2	0.2
AR coefficient gov. spending	ρ_χ	0.927	0.924	0.900	0.949	B	0.5	0.1
Gov. spending output coefficient	$\eta_{\chi,y}$	0.099	0.095	-0.230	0.412	N	0.1	0.2
AR coefficient short-run exp.	ρ_{eS}	0.606	0.601	0.529	0.677	B	0.2	0.05
Short-run exp. output coefficient	η_{eS}	0.151	0.139	0.038	0.247	N	0.1	0.2
Steady state growth rate	$100\bar{z}_a$	0.410	0.413	0.338	0.491	N	0.50	0.05
Steady state inflation	$100\left(\bar{\Pi}-1\right)$	0.721	0.720	0.639	0.802	N	0.75	0.05
AR coefficient preference	ρ_d	0.980	0.979	0.967	0.991	B	0.5	0.1
AR coefficient investment	ρ_i	0.446	0.463	0.337	0.589	B	0.5	0.1
AR coefficient long-run growth	ρ_a	0.223	0.236	0.161	0.309	B	0.5	0.1
AR coefficient repression	ρ_Γ	0.995	0.993	0.989	0.997	B	0.5	0.1
AR coefficient term premium	ρ_{tp}	0.089	0.098	0.033	0.160	B	0.5	0.1
AR coefficient bond holdings	ρ_b	0.977	0.975	0.964	0.986	B	0.5	0.1
S.d. preference	$100\sigma_d$	1.838	1.899	1.638	2.142	IG	0.5	2.0
S.d. investment	$100\sigma_i$	1.882	1.874	1.578	2.174	IG	0.5	2.0
S.d. monetary policy shock	$100\sigma_m$	0.190	0.193	0.168	0.215	IG	0.5	2.0
S.d. government spending ratio	$100\sigma_\chi$	3.491	3.547	3.135	3.962	IG	0.5	2.0
S.d. short-run expenditures	$100\sigma_{eS}$	0.784	0.801	0.696	0.913	IG	0.5	2.0
S.d. term premium shock	$100\sigma_{tp}$	4.101	4.100	3.339	4.822	IG	0.5	2.0
S.d. tax rate	$100\sigma_\tau$	0.476	0.486	0.426	0.544	IG	0.5	2.0
S.d. long-run growth	$100\sigma_a$	1.782	1.833	1.514	2.127	IG	0.5	2.0
S.d. repression	$100\sigma_\Gamma$	1.710	1.754	1.545	1.947	IG	0.5	2.0
S.d. bond holdings	$100\sigma_b$	4.981	5.245	3.844	6.536	IG	0.5	2.0
Log data density	3181.038							

Note: Posterior statistics based on MCMC with two chains of 400k draws each, 80% of draws discarded, and an acceptance rate of approx. 23%. The right part of the table reports the priors. In the column "Type", G, N, B, and IG correspond to gamma, normal, beta, and inverse gamma distributions, respectively.

in favor of Regime F when estimating a model that allows policy parameters to vary, thereby providing additional identification. We also verify below that our main result is robust across regimes.[9]

We compute additional results regarding the role of shocks, reporting conditional variance decompositions and historical decompositions in the online appendix. Here we note that the model implies that the repression shock fully explains the evolution of banks' debt holdings relative to total assets, shown in Figure 2 (see again equation (20)). However, the repression shock also accounts for around 18% and 20% of the variance of the debt-to-GDP ratio and banks' holdings of government debt relative to total debt outstanding, respectively. Its contribution to the variance of other variables is smaller. Regarding the historical role of repression shocks, initial conditions matter strongly for the decomposition in our analysis, for example because the debt-to-GDP ratio peaks at the start of our estimation sample. We investigate this issue in more detail in Section IV. To set the stage, we first inspect the mechanism at the heart of the model.

III. Inspecting the mechanism

In this section, to shed light on the mechanism through which financial repression affects the economy, we first zoom in on the market for government debt. Subsequently, we use model simulations to show that its impact on the macroeconomy is ambiguous—depending on the model parameters, it may either lower or raise economic activity and, in fact, the debt-to-GDP ratio.

A. Partial equilibrium

In what follows we develop some intuition for how financial repression impacts public finances in particular and the economy in general. In a first step, we take a partial equilibrium perspective and, to keep things simple, assume that debt is held exclusively by banks: $b_t^b = b_t$. Further, we abstract from inflation and assume a constant fiscal surplus f. A simplified version of the government budget constraint then reads as follows:

$$Q_t^b b_t = \frac{1 + \iota Q_t^b}{Q_{t-1}^b} Q_{t-1}^b b_{t-1} - f. \tag{53}$$

This expression implicitly defines the supply curve of government debt: it relates the current price of debt Q_t^b to the quantity of bonds b_t, given outstanding liabilities and bond prices in the previous period. The supply curve is downward sloping because a higher bond price reduces the amount of debt which needs to

to a Bayes factor of approximately 13.2, which, according to Jeffreys (1961) and Kass and Raftery (1995), is classified as "strong" evidence on the Jeffreys scale, though it does not reach the "decisive" threshold.

[9]The online appendix reports the estimates for Regime M and provides plots of the posterior distributions for both regimes.

be placed with banks in order to redeem a given amount of outstanding debt net of the surplus. We depict the supply curve as the blue solid line in Figure 3. It is labeled "S."

The same figure also features a demand curve for government debt, labeled "D", that determines the demand for government debt in the absence of repression or, equivalently, in case the regulatory constraint is not binding. Without loss of generality we assume it to be horizontal. As $R_t^b = (1 + \iota Q_t^b)/Q_{t-1}^b$, it is implicitly determined by banks' optimality condition (22) that ties the return on government debt to the deposit rate. The deposit rate, in turn, is proportional to the time-discount factor thanks to optimality condition (12). The intersection of "D" and "S" in Figure 3 determines the "laissez-faire" price $\tilde{Q}^b$ of debt that prevails in the absence of repression.

For a given market value of private securities held by the bank, $Q_t^s s_t^b$, the regulatory constraint (19) defines a downward sloping relationship between the price of debt and the amount of debt that needs to be held by banks whenever the regulatory constraint binds:

$$Q_t^b b_t^b \geq \Gamma_t Q_t^s s_t^b. \tag{54}$$

It is shown as a hyperbola in Figure 3, depicted in red and labeled RC. Intuitively, a binding constraint implies that either the volume held by banks is high and the price is low, or vice versa.

What determines the equilibrium in the market for government debt? Since we assume that the regulatory constraint binds, the equilibrium price Q^b is given by the intersection of RC and S. It exceeds the laissez-faire price. This price is consistent with the demand curve, because in case of repression the demand for government debt shifts upward in Panel A (from D to D'). Formally, this is brought about by a positive realization of the Lagrange multiplier μ in the bankers' optimality condition equation (22). Because holding an additional unit of government debt provides additional value to the bank if the regulatory constraint binds, bankers are ready to purchase government debt at a price which exceeds the laissez-faire price. Equivalently, for a given price of government debt in the next period, repression lowers the yield on government debt, as observed by Reinhart and Sbrancia (2015) and Acalin and Ball (2026), and the government needs to issue less debt in order to meet a given financing requirement.

A complementary perspective is that financial repression increases fiscal space, as stressed by Jeanne (2023). Panel B of Figure 3 illustrates this, by showing that repression allows the government to issue a larger amount of debt at the laissez-faire price. This is consistent with the observation by Chari, Dovis and Kehoe (2020) that governments are likely to "engage in financial repression when their fiscal needs are exceptionally high, such as during wartime." Concretely, in the right panel, the outward shift from S to S' reflects an increase in fiscal financing needs; repression, operating via the regulatory constraint on banks'

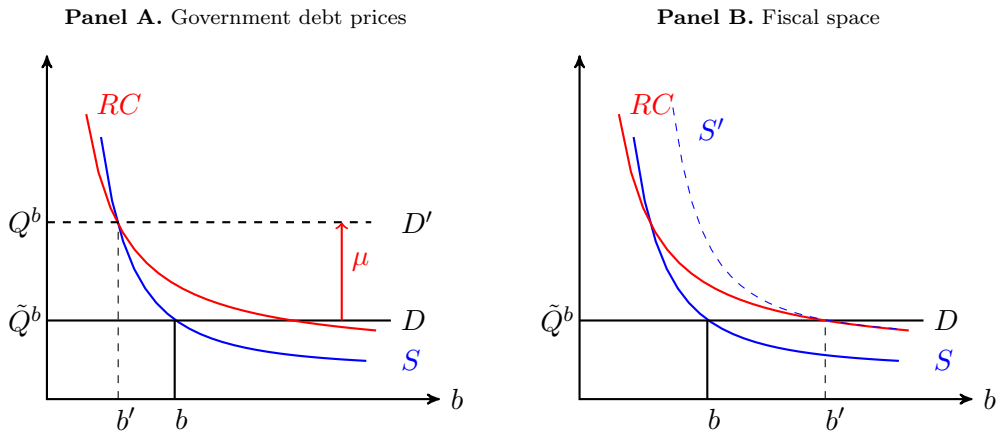


FIGURE 3. THE MARKET FOR GOVERNMENT DEBT

Note: Stylized representation of the market for government debt. Supply (demand) of debt is represented by the blue (black) line. Regulatory constraint represented by the RC curve. $\tilde{Q}^b$ is the laissez-faire price of debt. Q^b is the actual price. Repression either shifts demand for government debt upward (left panel) or supply outward (right panel).

demand, allows this. For instance, during WW2, the U.S. debt-to-GDP ratio rose by around 60 percentage points and the additional debt was placed with the private sector, and commercial banks in particular (Martin and Younger, 2020).

Financial repression, however, is not confined to the regulatory constraint that operates in our model. Other policy measures which may be understood as financial repression include yield-curve control and specific regulatory changes by the Federal Reserve (Alfriend, 1988; Garbade, 2020).[10] While these measures differ along specific dimensions, the regulatory constraint in our model is meant to capture their common effect—namely, reducing the effective interest rate on government debt—and thus allows us to assess the effects of financial repression in a quantitatively meaningful way.

At the same time, we acknowledge that forces other than financial repression or government regulation may generate a "funding advantage" of government debt over private assets. In particular, if government debt is perceived as especially safe and liquid, investors may want to hold a fraction of their portfolios in such securities. Under this interpretation, the resulting regulatory discount that emerges in our model is closely related to the convenience yield (Krishnamurthy and Vissing-Jorgensen, 2012; Payne and Szőke, 2025; Jiang, Krishnamurthy and Lustig, 2023); see online appendix for further details. In our analysis, the Treasury-demand shock—albeit only in reduced form—may capture some of these effects arising

[10]Between 1942 and 1972, the FED treated government bond holdings by banks as "riskless" to avoid penalizing them for holding these low-yield bonds. However, as shown by Fueki, Hürtgen and Walker (2024) and Abad (2019), zero-risk weights for government bonds lead to capital misallocation and constrain the overall supply of credit.

from households' Treasury demand. Consistent with this conjecture, we present evidence in Section IV that the repression shocks identified by our model are closely linked to the historical narrative on financial repression.

B. General equilibrium

We now zoom out and take a general equilibrium perspective to examine how financial repression affects the economy as a whole. In particular, we show that the model is not hard-wired to predict a specific change in the debt-to-GDP ratio following a repression shock. To do so, we compute impulse responses to a repression shock under the baseline parameterization, evaluated at the posterior mode, assuming that the economy is initially in steady state when the regulatory constraint temporarily tightens, i.e., Γ_t increases.

Figure 4 reports impulse responses for two parameterizations, both of which imply Regime F with respect to the monetary–fiscal policy mix. In each panel, the horizontal axis measures time in quarters and the vertical axis the deviation of each variable from its steady state. The solid (black) line shows results for the baseline estimates at the posterior mode. The shock itself is shown in Panel A.[11] It requires banks to increase their holdings of government debt relative to their total assets. As banks' demand for government debt increases, its price rises, as discussed above. This, in turn, lowers the expected return relative to the deposit rate under both parameterizations, as shown in Panel B. At the same time, increased repression forces banks to curtail lending to firms, such that the expected return on capital (relative to the deposit rate) increases in the baseline. This is shown in Panel C of the figure.[12]

We contrast this finding with the dashed (blue) line, which shows results under an alternative specification with greater price stickiness, more counter-cyclical fiscal policy, and more accommodative monetary policy.[13] Under the alternative parameterization, the expected return on capital falls relative to the deposit rate, reflecting different macroeconomic dynamics, shown in the bottom panels of Figure 4. In the baseline, the repression shock is recessionary. Higher funding costs reduce investment (Panel D), as well as output (not shown) and inflation (Panel E) amid weaker demand. In contrast, the shock is expansionary in the alternative scenario because the fiscal–monetary policy mix is more accommodating and also more potent in the presence of greater price stickiness. Higher inflation leads to an increase in the short-term interest rate, lowering the expected return on capital relative to the deposit rate and thereby further stimulating investment. Banks' demand for government bonds rises, leading to an even stronger decline in

[11] Our estimation implies that the shock is very persistent. Here we assume a more temporary shock ($\rho_\Gamma = 0.9$) to make the dynamics more transparent.

[12] The online appendix reports the impulse responses to a repression shock under Regime M. The qualitative pattern is similar to the baseline.

[13] In the baseline estimates, the response of tax revenues to output is pro-cyclical ($\eta_{\tau,y} > 0$). In the alternative scenario, we assume it is counter-cyclical. For details on the parameters see the notes to the figure.

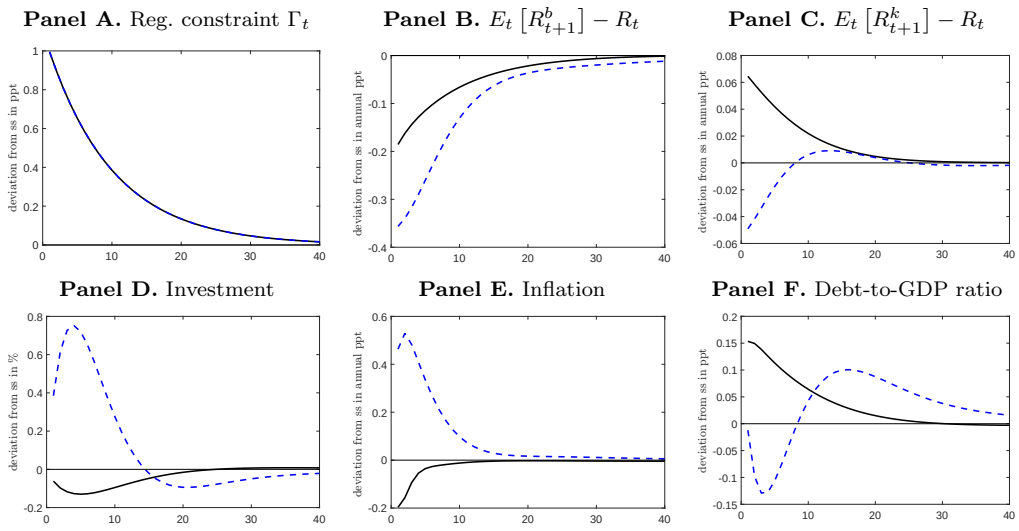


FIGURE 4. IMPULSE RESPONSES TO A REPRESSION SHOCK

Note: Repression shock is a temporary increase of Γ_t by one standard deviation. Solid lines show shock responses for baseline estimates at posterior mode, see Table 2 below. Dashed (blue) lines show responses under alternative scenario with policy parameters (responsiveness of tax revenues to GDP, $\eta_{\tau,y}$: -0.50 instead of 0.12; responsiveness of policy rate to inflation, η_π: 0.425 instead of 0.72) and increased price rigidity (γ_p: 0.85 instead of 0.55). The vertical axis measures deviation from steady state, while the horizontal axis measures time in quarters.

the expected return on government debt relative to the baseline (Panel B). Eventually, these differences matter for the response of the debt-to-GDP ratio to the repression shock: it rises in the baseline but declines in the alternative scenario, where nominal GDP—the denominator—increases more than government debt.

Hence, depending on parameter values, the model allows for markedly different dynamics following a repression shock. We demonstrate this systematically using a prior-predictive analysis following Geweke (2010); see online appendix for details. The prior-implied distribution of impulse responses spans a wide range of outcomes, implying that the increase in the debt-to-GDP ratio in the baseline is data-driven rather than a mechanical implication of the model.

The simulations also show that, consistent with its interpretation as a form of capital income taxation emphasized by CDK, financial repression distorts investment decisions. By contrast, taxes in the model are assumed to be lump sum. One may therefore wonder whether the model stacks the odds against repression as a source of government funding. To address this concern, we consider a version of the model with distortionary labor taxes, but find that the adjustment to a repression shock is essentially unchanged (see online appendix). To understand this result, note that taxes evolve according to feedback rule (45), rather than being adjusted to balance the budget. As a result, any funding advantage of the government—and the associated savings in interest payments—arising from

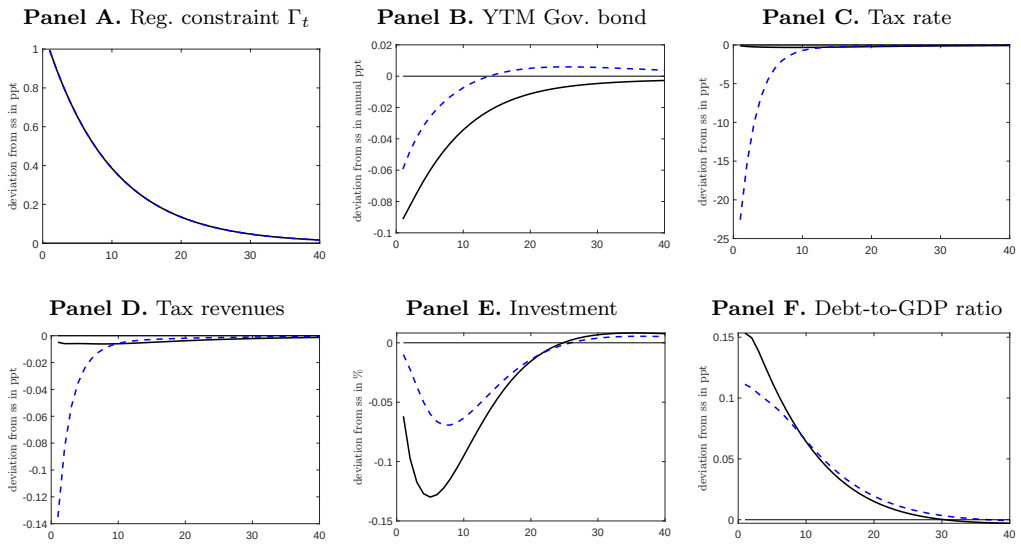


FIGURE 5. IMPULSE RESPONSES TO A REPRESSION SHOCK W/O AND W/ TAX REBATE

Note: Repression shock is a temporary increase of Γ_t by one standard deviation. Solid (black) lines reproduce result for baseline model from Figure 4. Dashed (blue) lines represent alternative scenario in which interest rate savings are rebated to households via reduction in the labor tax rate. Vertical axis measures deviations from steady state, while horizontal axis measures time in quarters.

repression are not directly rebated via tax reductions. Instead, they affect the path of debt; consequently, whether taxes are lump sum or distortionary has little impact on the dynamics following a repression shock.

To further illustrate this point, we conduct an experiment in which we combine a repression shock with a tax shock that rebates the interest savings associated with repression directly to households, resulting in a temporary reduction in the labor tax rate. The dashed (blue) lines in Figure 5 show the results, contrasted with the baseline (solid black lines) reproduced from Figure 4. Panel B shows the decline in the interest rate due to repression, measured by the yield to maturity on a ten-year government bond. This generates interest savings that are used to lower the labor tax rate. Specifically, the labor tax rate is reduced for about two years (Panel C).[14] Tax revenues fall accordingly (Panel D), but the tax cut stimulates investment, which declines less than in the baseline (Panel E), as well as GDP (not shown), and, in turn, the debt ratio rises less strongly than in the baseline (Panel F). Overall, this experiment supports the notion that, if interest rate savings were used to reduce other distortions in the economy, the adverse effects of a repression shock would be less severe than in the baseline.

[14]We scale the labor tax shock such that it induces an increase in interest payments, $R_t^b \cdot b_{t-1}$, that exactly offsets the impact of the repression shock over a 10-year horizon.

IV. External validation

In this section, we evaluate the model's performance along several dimensions. First, we quantify the contribution of repression shocks to the evolution of the debt-to-GDP ratio and relate the identified shocks to historical narratives on financial repression. Second, we compare the model predictions for time series not used in the estimation with their empirical counterparts. Finally, we estimate the responses of additional variables—also not used in the estimation—to the repression shocks identified in the estimation in order to corroborate the mechanism underlying the model.

A. Historical decomposition and narratives

Based on a historical decomposition, the top panel in Figure 6 shows the contribution of the repression shock—indicated by the black bars—to the evolution of the debt-to-GDP ratio (solid line). The figure is based on an extended data set starting in 1936. We apply the Kalman filter evaluated at the posterior mode, based on estimates from our post–World War II sample. This approach avoids estimating the model on war-period data while still allowing us to assess the role of repression in the build-up of debt, which peaked during the war. Indeed, repression made a sizable contribution, accounting for roughly one quarter of the increase in debt according to our estimates.[15]

To assess the plausibility of these estimates, Panel B of Figure 6 relates the smoothed shock series to well-known policy measures that—by design or as a side effect—contributed to financial repression. In this context, moral suasion can be viewed as a form of repression policy. Already used during World War I—when Treasury and Federal Reserve officials organized Liberty Bond drives (Mullin, 2021)—it was revived during World War II as officials urged banks to convert excess reserves into government bonds (Rose, 2021).

The Banking Act of 1933 introduced Regulation Q, which prohibited interest on demand deposits and imposed ceilings on time and savings deposit rates, lowering banks' funding costs and facilitating the maintenance of low Treasury yields (Reinhart and Sbrancia, 2015). During the war, banks absorbed the bulk of newly issued federal debt (Garbade, 2020). They were incentivized to do so because government securities were exempted from the computation of the "equity-to-risky-asset ratio" (Mitchell, 1984; Alfriend, 1988). From mid-1942 onward, the Treasury yield curve was effectively pegged, with bills fixed at 3/8 percent and long-term bonds near 2.5 percent to "encourage prompt buying of securities by investors, who might otherwise have awaited higher rates" and to "keep down the interest cost on the Government's war debt" (Thomas and Young, 1947).

Turning to the postwar period, we first note that until the 1970s capital standards and risk-asset approaches were continuously adjusted and refined by the

[15]The online appendix reports the historical decomposition for all observables.

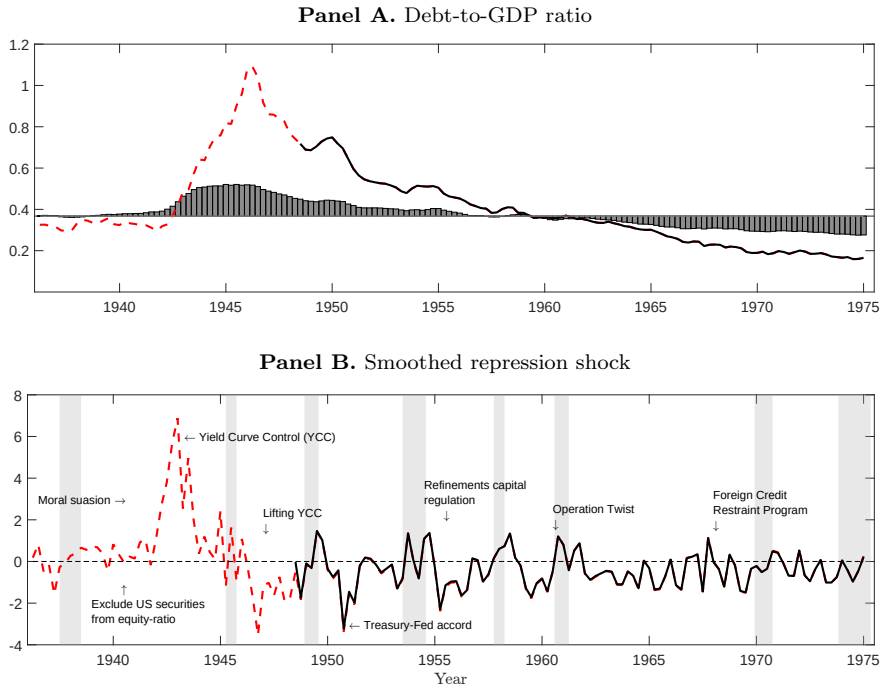


FIGURE 6. REPRESSION SHOCK AND ITS HISTORICAL CONTRIBUTION

Note: Top panel shows contribution of repression shocks to debt-to-GDP ratio (black bars) based on a historical decomposition; bottom panel shows the smoothed shock series $\epsilon_{\Gamma,t}$ together with NBER recessions (shaded areas). In both panels, the black line relates to the estimation sample, the red dashed line to the extended sample starting in 1936.

three federal regulatory agencies—the Federal Reserve System (Fed), the Federal Deposit Insurance Corporation (FDIC), and the Office of the Comptroller of the Currency (OCC)—which oversee commercial banks (see Alfriend, 1988). In parallel, policies affecting the pricing of government debt directly were gradually relaxed. The Federal Reserve ended the 3/8 percent bill-buying rate in 1947 and later ceased defending yield ceilings on some issues, culminating in the 1951 Treasury–Fed Accord. Consistent with this historical record, Panel B of Figure 6 shows that the contribution of repression shocks to the build-up of debt turns negative in the late 1950s.

That said, Operation Twist stands out in our shock measure—an initiative to reduce long-term interest rates while maintaining or raising short-term rates to support the balance of payments under Bretton Woods. From a financial-repression perspective, it is noteworthy that official balance-sheet operations that compressed long-term yields worked in tandem with deposit-rate ceilings and supervisory preferences that channeled bank portfolios into government debt at administratively low returns. Likewise, the 1968 Foreign Credit Restraint Program also appears prominently in our shock series.

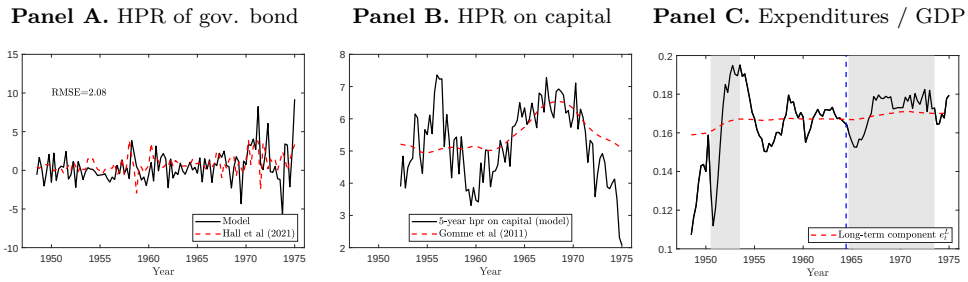


FIGURE 7. EXTERNAL VALIDATION OF MODEL PREDICTIONS

Note: Panel A shows nominal holding period return (HPR) of a government bond portfolio implied by the model and estimates by Hall et al. (2021). Panel B shows the 5-year average holding period return on capital implied by the model and corresponding estimates by Gomme, Ravikumar and Rupert (2011). Panel C shows government expenditure-to-GDP ratio and e_t^L its long-term component (red-dashed). The shaded areas are the Korean and Vietnam wars and the blue vertical line corresponds to president Lyndon B. Johnson's first announcement of the "Great Society."

Overall, our estimates are consistent with the historical record suggesting that repression was gradually phased out from the late 1950s onward.

B. Further time series evidence

Figure 7 compares the model's predictions for time series that were not used in the estimation with the corresponding data. In Panel A, we compare the ex-post nominal return on the average-maturity government debt portfolio (holding-period return) predicted by the model with an empirical measure compiled by Hall et al. (2021).[16] We find that the two series co-move closely.

In Panel B of the same figure, we compare the time series for the return on capital compiled by Gomme, Ravikumar and Rupert (2011) with the model-implied five-year holding-period return on capital. In the model, firms do not issue marketable bonds. Hence, we follow Gertler and Karadi (2013) and construct a synthetic 5-year asset whose cash flow matches the model's return on capital. At lower frequencies, the two series align fairly well. This comparison is informative because the real return on private assets helps gauge the government's funding advantage relative to the private sector. In the model, this advantage increases with financial repression—consistent with recent evidence by Payne and Szőke (2025) on the role of bank regulation for the funding advantage. Details on the computation of the five-year holding-period return are provided in the online appendix, where we also discuss the extent to which financial repression can account for the convenience yield, particularly in light of recent work showing that the government's funding advantage does not exhibit a stable relationship

[16]They use market prices for all marketable public bonds and compute the ex-post holding-period return. This measure captures changes in bond valuations that are not reflected in interest expenses calculated from coupon payments and is therefore an appropriate benchmark for the model prediction.

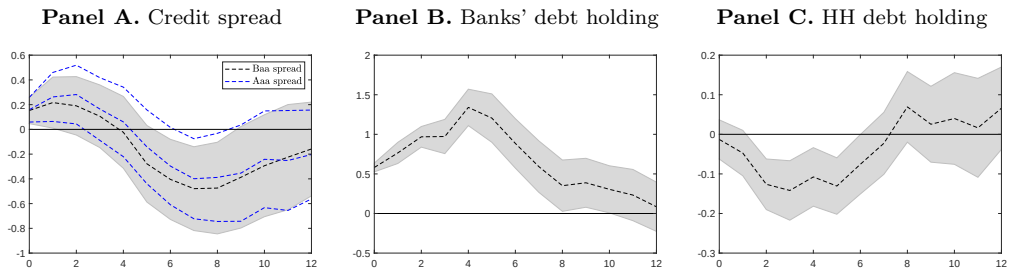


FIGURE 8. RESPONSES TO REPRESSION SHOCKS

Note: Impulse responses to repression shock estimated using local projections; area indicates 16% and 84% probability bands. Repression shock identified in estimated model. Data on spreads and debt holdings not used in estimation; see online appendix for details.

with the debt-to-GDP ratio (Krishnamurthy and Vissing-Jorgensen, 2012; Lehner et al., 2025).

Finally, in Panel C, we show government expenditures as a ratio of GDP (e_t) and their long-term component (e_t^L), as estimated by the model. We see that expenditures are particularly high during the Korean and Vietnam Wars, highlighted by the gray areas in the figure. Consistent with the findings of Bianchi and Ilut (2017), the long-term component e_t^L rises following the first announcement of the Great Society by President Lyndon B. Johnson in May 1964. In light of this, we consider the estimates for e_t^L plausible: they capture large expenditure programs that arise from political processes not explicitly modeled here.

C. Additional evidence on the effects of repression shocks

Figure 4 above illustrates that a repression shock raises excess returns on capital in the baseline. In that case, the mechanism in the model implies higher expected excess returns on other securities as well. To validate this prediction, we estimate impulse responses of the Aaa and Baa spreads over the short-term interest rate to the identified repression shocks from the estimated model. The left panel of Figure 8 shows that both spreads increase, as suggested by the baseline, even though they were not used in the estimation. The online appendix provides details on the data and the estimation.

Similarly, the model has implications for quantities that we assess using data not employed in the estimation in order to provide further external validation. In particular, we use local projections to estimate the dynamic adjustment of government debt holdings of banks and households using data from the U.S. Financial Accounts. Consistent with the model, Panel B of Figure 8 shows that, in response to a repression shock, banks' holdings of government debt increase.[17] Given

[17]This measure of bank holdings is somewhat broader than the one used in the estimation of the model; see online appendix for details.

the mechanism in the model, we would expect unconstrained investors—such as households—to reduce their government bond holdings when yields fall. The result in Panel C confirms this conjecture, too.[18]

V. Financial repression in the U.S., 1948–1974

We are finally in a position to address the question that motivates our analysis: did financial repression affect how the debt-to-GDP ratio evolved during our sample period, and, if so, through which channels?

A. What if? A general-equilibrium counterfactual

We proceed by constructing a counterfactual based on the estimated model. Our starting point is the observation that, according to our estimates, financial repression declined steadily over the sample period. Consistent with the historical narrative discussed above, the top-left panel of Figure 9 shows that the estimate for Γ_t declines from about 0.9 at the beginning of the sample to nearly zero by 1975. Against this background, we first consider a counterfactual in which financial repression is not phased out. We also discuss an alternative scenario in which repression is fully removed at the beginning of the sample. Both counterfactuals yield similar insights, but the first—persistently high repression—is easier to implement as it is nested in the baseline. Note that both counterfactuals start from the same 1948q2 initial conditions and hence hold fixed the debt level that earlier repression helped to accumulate.

We contrast the results for a scenario of constantly high repression with the actual outcomes in the panels of Figure 9: the solid (black) line represents the outcome according to the estimated model (that is, our baseline), the dashed (red) line shows the counterfactual outcome in case repression is not phased out. This is reflected in Panel A of Figure 9 by a value for Γ_t which is constantly high during our sample period.

Panel B reports the central result: the debt-to-GDP ratio would have declined less had financial repression not been phased out. As discussed in Section III.B, the model does not mechanically predict the effect of repression on the debt ratio. In the estimated model, however, repression raises it. The counterfactual mirrors the impulse-response analysis. Panel C shows that banks' holdings of government debt rise markedly in the counterfactual, as repression forces a portfolio shift toward public debt. This crowds out private investment: although investment increases over time in both cases, it is persistently lower under high repression (Panel D). As a result, GDP is also lower (Panel E). The fiscal implications follow

[18] Furthermore, banking regulation may be understood as a form of financial repression: A tightening of repression is akin to stricter capital or liquidity requirements that favor government debt over private lending. Consistent with this mechanism, Eickmeier, Kolb and Prieto (2023) and Drechsel and Miura (2025) find, using post-1980s data, that tighter banking regulation widens credit spreads, contracts credit supply, and reduces real activity.

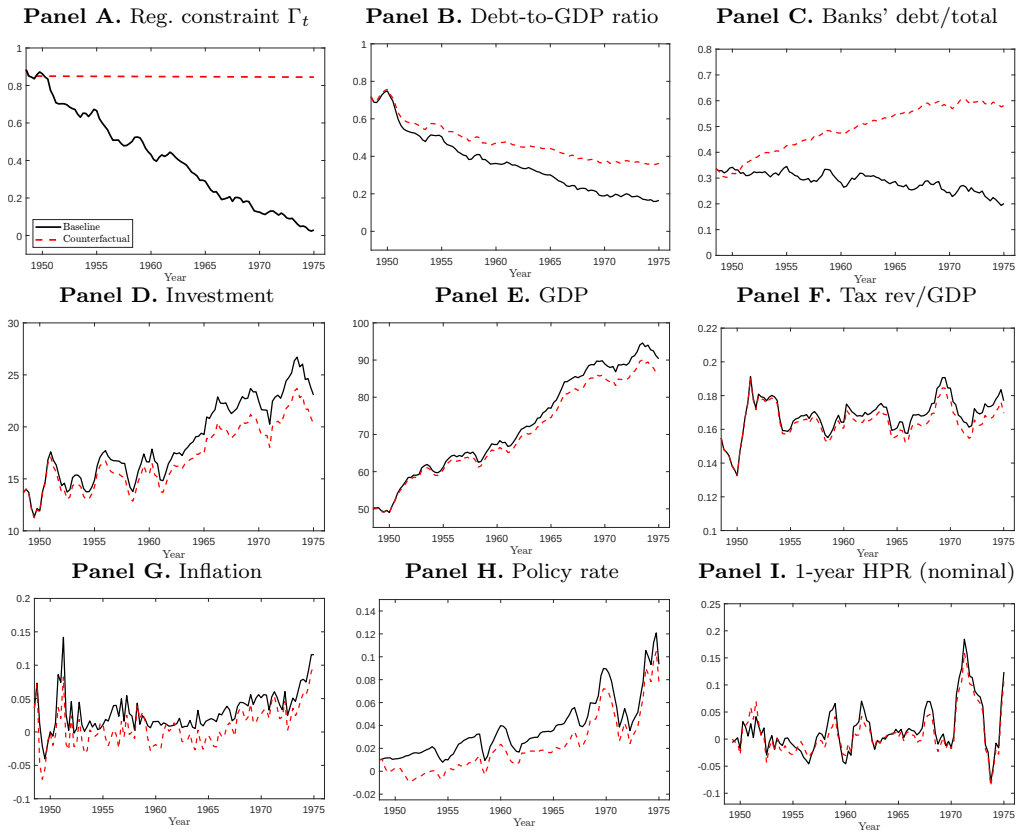


FIGURE 9. GENERAL-EQUILIBRIUM COUNTERFACTUAL

Note: Solid (black) lines represent actual outcomes, as predicted by the model at the posterior mode. Dashed (red) lines show results for counterfactual in which repression is kept high at the level observed at the beginning of the sample period; see Panel A.

directly. Tax revenues (Panel F) are lower when repression remains high, reflecting weaker economic activity. Together, these effects explain why the debt-to-GDP ratio declines less in the counterfactual.

As financial repression depresses economic activity, inflation is lower (Panel G), and monetary policy sets a lower policy rate (Panel H). Finally, the holding-period return (HPR) remains broadly similar across the baseline and the counterfactual. The mechanism is similar to that discussed earlier: in the baseline, the phasing-out of repression generates persistent capital losses as bond prices decline toward their laissez-faire level. Under sustained repression, bond prices remain elevated but yield no further capital gains. As a result, the HPR stays close to the baseline despite a lower ex ante yield.

To illustrate the quantitative importance of the effects of financial repression, Figure 10 zooms in on key variables, showing the difference between the counterfactual outcome, assuming that repression is kept in place, and the baseline,

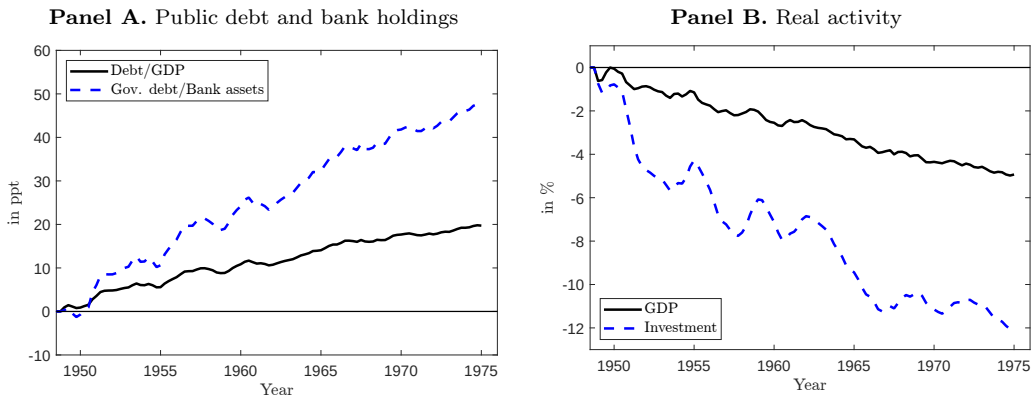


FIGURE 10. EFFECTS OF SUSTAINED FINANCIAL REPRESSION

Note: Difference for key variables across counterfactual outcome w/ repression and baseline w/ phasing out of repression. The figure shows the difference across outcomes shown in Figure 9.

where repression is phased out. Panel A shows the difference for the public debt ratio (solid black line) and the share of government debt in banks' assets (blue dashed line). Differences are large. In the counterfactual the ratio of public debt is 20 percentage points higher at the end of the sample period. Banks' holdings of government debt are 50 percentage points higher. Hence, we conclude that phasing out repression has actually *lowered* the public debt ratio significantly. To a large extent this is because phasing out repression provided a boost to real activity. In panel B we observe that investment would have been depressed by up to 12 percent, and output by up to five percent, had repression not been phased out.

Our counterfactual illustrates the effects of financial repression by contrasting a scenario with permanently high repression with the baseline, in which repression is phased out. A natural alternative is to consider a counterfactual in which repression is fully removed at the start of the sample period (1948:Q2). This scenario is more demanding conceptually, as the absence of repression is not nested in a specific parameter value. Nevertheless, we compute this counterfactual by removing the regulatory constraint altogether, so that the banking sector effectively reduces to the framework of Gertler and Karadi (2013). To ensure comparability, we initialize the simulation at the same starting values as in the baseline. The results confirm the robustness of our findings. When repression is removed immediately, banks reduce their holdings of government debt, nominal bond returns increase, investment expands, and both output and inflation rise. As a result, the debt-to-GDP ratio declines more rapidly in the short and medium run. We show these results in the online appendix.

B. An accounting perspective

Our results may appear to contradict the conventional view that financial repression helps reduce government debt by lowering interest rates. Indeed, influential work by Reinhart and Sbrancia (2015) and, more recently, Acalin and Ball (2026) argues that financial repression reduced the debt-to-GDP ratio in the U.S. after World War II. However, these studies take a narrower perspective, focusing on the direct effect of repression on interest rates. Our framework allows us to isolate this channel as well. Abstracting from general-equilibrium effects, we compute the laissez-faire interest rate that would prevail in the absence of repression and then trace the path of government debt assuming, all else equal, that the government had paid this rate rather than the observed one.

Formally, we define the laissez-faire rate on the basis of equation (22): by setting $\mu_{i,t}$ to zero at all times, we obtain the interest rate which would clear the market for government debt, assuming that banks were not constrained by financial repression. The laissez-faire rate, R^{LF}_{t+1}, is then implicitly defined by the following expression:

$$E_t M_{t,t+1} \left(R^{LF}_{t+1} - R^d_t \right) \frac{\Omega_{t+1}}{\Pi_{t+1}} = \frac{\lambda_t \theta \Delta}{1 + \lambda_t}. \tag{55}$$

Based on the ex ante laissez-faire rate we can also compute the laissez-faire HPR R^{LF}_t which enters the law of motion for the laissez-faire debt-to-GDP ratio, $\mathcal{B}^{LF}_t$, in our accounting counterfactual:

$$\mathcal{B}^{LF}_t = R^{LF}_t \frac{\mathcal{B}^{LF}_{t-1} y_{t-1}}{\Pi_t y_t} - \tau_t + e_t + z_{tp,t}. \tag{56}$$

Figure 11 shows the results, contrasting the baseline (solid black line) and the counterfactual laissez-faire outcome (dashed blue line). Panel A displays the 1-year HPR: in line with the results of Reinhart and Sbrancia (2015) and Acalin and Ball (2026), the laissez-faire HPR is considerably higher than what we obtain for the baseline. The gap lasts until the end of the 1960s, reflecting our assessment that, while repression was phased out during our sample period, there was non-trivial repression for most of the time—see again the estimate for Γ_t shown in Panel A of Figure 9.

Taken at face value, these estimates suggest that the government borrowed at reduced interest rates. How much did this contribute to the decline in public debt? To address this question in partial equilibrium, we compute the path of debt using equations (55) and (56); the results are shown in Panel B of Figure 11.[19] Over the sample period, the debt-to-GDP ratio declined by more than 55 percentage points (solid black line). At the higher laissez-faire rate, the decline

[19] Online appendix results for Regime M are very similar.

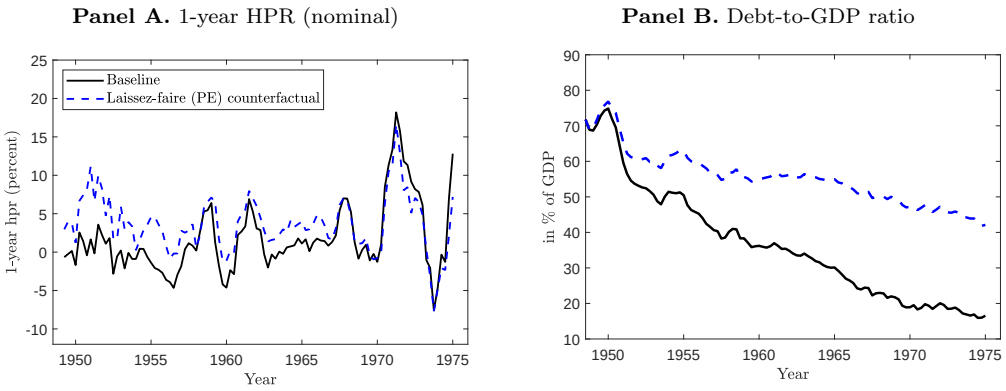


FIGURE 11. ACTUAL DEVELOPMENTS V PARTIAL-EQUILIBRIUM COUNTERFACTUAL

Note: The baseline is given by the solid (black) line and the laissez-faire counterpart by the dashed (blue) line. The laissez-faire scenario is based on a partial-equilibrium perspective: It obtains by setting $\mu_t = 0$ in eq. (22), all else equal.

would have been only about 30 percentage points (dashed blue line), implying that—*all else equal*—the debt ratio would have been roughly 26 percentage points higher by 1974. This is consistent with Acalin and Ball (2026), who attribute a 28 percentage point reduction in the debt-to-GDP ratio to "distorted interest rates."[20] Yet, in general equilibrium, all else is not equal.

VI. Conclusion

We document evidence of financial repression in the U.S. after WW2. At the same time, our estimated general equilibrium model suggests that repression was gradually phased out after the war, in line with narrative accounts. Did repression help liquidate public debt during this period? Contrary to earlier work, we find that it did not. This conclusion is based on a model-based counterfactual in which repression is assumed to persist rather than to be phased out. In that scenario, output and inflation would have been lower, chiefly because repression crowds out bank lending and depresses investment. Consequently, the debt-to-GDP ratio would have been significantly higher had repression remained at the levels observed immediately after the war.

At the same time, when focusing solely on the direct effect of financial repression on interest rates, repression does appear to have facilitated the reduction of government debt. This result is intuitive and may make repression attractive to policymakers. However, such a partial-equilibrium perspective misses the indirect adverse effects operating through general equilibrium, which we find to dominate over our sample period. Our analysis thus cautions against a continued

[20] Acalin and Ball (2026) focus on the pre-1951 period but view this estimate as a lower bound for the overall impact of financial repression during our sample.

reliance on financial repression as a debt-reduction tool. Note that our counterfactuals condition on the fiscal position inherited in 1948, which was itself shaped by wartime repression. Moreover, because our counterfactual starts only in 1948, after key elements of financial repression had already been introduced during World War II, it does not address how inflation, debt, and the broader economy would have evolved had repression been avoided altogether. Addressing this question requires modeling the wartime economy explicitly, which we leave for further research. What we do establish is that, from a position of high debt, maintaining repression would have been counterproductive to the very objective it is commonly associated with.

REFERENCES

Abad, Jorge. 2019. "Breaking the Sovereign-Bank Nexus." CEMFI Working Paper.

Acalin, Julien, and Laurence M. Ball. 2026. "Did the United States Really Grow Out of Its World War II Debt?" *American Economic Journal: Macroeconomics*, 18(3): 112–135.

Aiyagari, S. Rao, and Ellen R. McGrattan. 1998. "The optimum quantity of debt." *Journal of Monetary Economics*, 42(3): 447–469.

Alfriend, Malcolm C. 1988. "International Risk-Based Capital Standard: History and Explanation." *Federal Reserve Bank of Richmond Economic Review*, 74(6): 28–34.

Arslanalp, Serkan, and Barry Eichengreen. 2023. "Living with High Public Debt." In *Structural Shifts in the Global Economy*. Kansas City, MO:Federal Reserve Bank of Kansas City. Proceedings of the Jackson Hole Economic Policy Symposium.

Barro, Robert J. 1979. "On the Determination of the Public Debt." *Journal of Political Economy*, 87(5): 940–971.

Barro, Robert J., and Francesco Bianchi. 2026. "Fiscal Influences on Inflation in OECD Countries, 2020–2023." *The Economic Journal*, 136(674): 626–654.

Becker, Bo, and Victoria Ivashina. 2018. "Financial Repression in the European Sovereign Debt Crisis." *Review of Finance*, 22(1): 83–115.

Bianchi, Francesco, and Cosmin Ilut. 2017. "Monetary/Fiscal Policy Mix and Agents' Beliefs." *Review of Economic Dynamics*, 26: 113–139.

Brock, Philip L. 1989. "Reserve Requirements and the Inflation Tax." *Journal of Money, Credit and Banking*, 21(1): 106–121.

Calvo, Guillermo A. 1983. "Staggered prices in a utility-maximizing framework." *Journal of Monetary Economics*, 12(3): 383–398.

Chandler, Lester V. 1949. "Federal Reserve Policy and the Federal Debt." *The American Economic Review*, 39(2): 405–429.

Chari, V. V., Alessandro Dovis, and Patrick J. Kehoe. 2020. "On the Optimality of Financial Repression." *Journal of Political Economy*, 128(2): 710–739.

Christiano, Lawrence J., Martin Eichenbaum, and Charles L. Evans. 2005. "Nominal Rigidities and the Dynamic Effects of a Shock to Monetary Policy." *Journal of Political Economy*, 113(1): 1–45.

Coenen, Günter, Peter Karadi, Sebastian Schmidt, and Anders Warne. 2018. "The New Area-Wide Model II: an extended version of the ECB's microfounded model for forecasting and policy analysis with a financial sector." European Central Bank Working Paper Series 2200.

Del Negro, Marco, Domenico Giannone, Marc P. Giannoni, and Andrea Tambalotti. 2017. "Safety, Liquidity, and the Natural Rate of Interest." *Brookings Papers on Economic Activity*, 48(1): 235–316. Spring issue.

Drechsel, Thomas, and Ko Miura. 2025. "The macroeconomic effects of bank regulation: New evidence from a high-frequency approach." University of Maryland.

Du, Wenxin, Alexander Tepper, and Adrien Verdelhan. 2018. "Deviations from Covered Interest Parity." *Journal of Finance*, 73(3): 915–957.

Eickmeier, Sandra, Benedikt Kolb, and Esteban Prieto. 2023. "The Macroeconomic Effects of Bank Capital Regulation." Centre for Economic Policy Research CEPR Discussion Paper DP18404. Published 29 August 2023.

Fry, Maxwell. 1997. "In Favour of Financial Liberalisation." *The Economic Journal*, 107(442): 754–770.

Fueki, Takuji, Patrick Hürtgen, and Todd B. Walker. 2024. "Zero-risk weights and capital misallocation." *Journal of Financial Stability*, 72: 101264.

Garbade, Kenneth D. 2020. "Managing the Treasury Yield Curve in the 1940s." Federal Reserve Bank of New York Staff Reports 913.

Gertler, Mark, and Peter Karadi. 2011. "A model of unconventional monetary policy." *Journal of Monetary Economics*, 58(1): 17–34.

Gertler, Mark, and Peter Karadi. 2013. "QE 1 vs. 2 vs. 3. . . : A Framework for Analyzing Large-Scale Asset Purchases as a Monetary Policy Tool." *International Journal of Central Banking*, 9(1): 5–53.

Geweke, John. 1999. "Using Simulation Methods for Bayesian Econometric Models: Inference, Development, and Communication." *Econometric Reviews*, 18(1): 1–73.

Geweke, John. 2010. *Complete and Incomplete Econometric Models.* Princeton University Press.

Giovannini, Alberto, and Martha de Melo. 1993. "Government Revenue from Financial Repression." *The American Economic Review*, 83(4): 953–963.

Gomme, Paul, B. Ravikumar, and Peter Rupert. 2011. "The return to capital and the business cycle." *Review of Economic Dynamics*, 14(2): 262–278.

Hall, George J., and Thomas J. Sargent. 2011. "Interest Rate Risk and Other Determinants of Post-WWII US Government Debt/GDP Dynamics." *American Economic Journal: Macroeconomics*, 3(3): 192–214.

Hall, George J., and Thomas J. Sargent. 2021. "Debt and taxes in eight U.S. wars and two insurrections." In *The Handbook of Historical Economics.* , ed. Alberto Bisin and Giovanni Federico, 825–880. Academic Press.

Hall, George J., and Thomas J. Sargent. 2022. "Three World Wars: Fiscal–Monetary Consequences." *Proceedings of the National Academy of Sciences*, 119(18): e2200349119.

Hall, George J., Jonathan Payne, Thomas J. Sargent, and Bálint Szőke. 2021. "US Federal Debt 1776–1960: Quantities and Prices." New York University mimeo.

Hetzel, Robert L., and Ralph F. Leach. 2001*a*. "After the Accord: Reminiscences on the Birth of the Modern Fed." *Federal Reserve Bank of Richmond Economic Quarterly*, 87(1): 57–64.

Hetzel, Robert L., and Ralph F. Leach. 2001*b*. "The Treasury-Fed Accord: A New Narrative Account." *Federal Reserve Bank of Richmond Economic Quarterly*, 87(1): 33–55.

Ilut, Cosmin L., Pavel Krivenko, and Martin Schneider. 2024. "Uncertainty or Frictions? A Quantitative Model of Scarce Safe Assets." National Bureau of Economic Research NBER Working Paper 32198.

IMF. 2023. "World Economic Outlook: Navigating Global Divergences." International Monetary Fund World Economic Outlook, Washington, DC.

Jeanne, Olivier. 2023. "'Whatever It Takes': Government Default Versus Financial Repression." Mimeo, Johns Hopkins University.

Jeffreys, Harold. 1961. *Theory of Probability.* . 3 ed., Oxford:Oxford University Press.

Jiang, Zhengyang, Arvind Krishnamurthy, and Hanno Lustig. 2023. "Implications of Asset Market Data for Equilibrium Models of Exchange Rates." National Bureau of Economic Research Working Paper 31851.

Kass, Robert E., and Adrian E. Raftery. 1995. "Bayes factors." *Journal of the American Statistical Association*, 90(430): 773–795.

Krishnamurthy, Arvind, and Annette Vissing-Jorgensen. 2012. "The Aggregate Demand for Treasury Debt." *Journal of Political Economy*, 120(2): 233–267.

Leeper, Eric M. 1991. "Equilibria under 'active' and 'passive' monetary and fiscal policies." *Journal of Monetary Economics*, 27(1): 129–147.

Leeper, Eric M., Nora Traum, and Todd B. Walker. 2017. "Clearing Up the Fiscal Multiplier Morass." *American Economic Review*, 107(8): 2409–2454.

Lehner, Clemens, Jonathan Payne, Jack Shurtleff, and Bálint Szőke. 2025. "The US Treasury Funding Advantage Since 1860." Princeton University Discussion Paper.

Martin, Antoine, and Joshua Younger. 2020. "War Finance and Bank Leverage: Lessons from History."

McKinnon, Ronald R. 1973. *Money and Capital in Economic Development.* Brookings Institution Press.

Mian, Atif, Ludwig Straub, and Amir Sufi. 2025. "A Goldilocks Theory of Fiscal Deficits." *American Economic Review*, 115(12): 4253–4291.

Mitchell, Karlyn. 1984. "Capital Adequacy at Commercial Banks." *Federal Reserve Bank of Kansas City Economic Review*, 69(September): 17–30.

Mullin, John. 2021. "A Look Back at Financial Repression." *Federal Reserve Bank of Richmond Econ Focus*, 22–25.

Ongena, Steven, Alexander Popov, and Neeltje Van Horen. 2019. "The Invisible Hand of the Government: Moral Suasion during the European Sovereign Debt Crisis." *American Economic Journal: Macroeconomics*, 11(4): 346–379.

Payne, Jonathan, and Bálint Szőke. 2025. "The Fragility of Government Funding Advantage." Princeton University Discussion Paper.

Reinhart, Carmen M., and M. Belen Sbrancia. 2015. "The liquidation of government debt." *Economic Policy*, 30(82): 291–333.

Reinhart, Carmen M., Vincent Reinhart, and Kenneth Rogoff. 2015. "Dealing with debt." *Journal of International Economics*, 96: S43 – S55.

Rose, Jonathan. 2021. "Yield Curve Control in the United States, 1942 to 1951." *Federal Reserve Bank of Chicago Economic Perspectives*, 1–16. No. 2.

Roubini, Nouriel, and Xavier Sala-i-Martin. 1995. "A growth model of inflation, tax evasion, and financial repression." *Journal of Monetary Economics*, 35(2): 275 – 301.

Shaw, Edward S. 1973. *Financial Deepening in Economic Development.* Oxford University Press.

Sims, Eric, and Jing Cynthia Wu. 2021. "Evaluating Central Banks' tool kit: Past, present, and future." *Journal of Monetary Economics*, 118(C): 135–160.

Thomas, Woodlief, and Ralph A. Young. 1947. "Problems of Post-War Monetary Policy." In *Postwar Economic Studies No. 8: Federal Reserve Policy.* Washington, DC:Board of Governors of the Federal Reserve System.

Woodford, Michael. 2001. "Fiscal Requirements for Price Stability." *Journal of Money, Credit and Banking*, 33(3): 669–728.